34721/34722 Linear Control Design 1
Spring Semester

Teachers: Silvia Tolu - Dimitrios Papageorgiou

Lesson 6/13 – Bode plots and Stability

Schedule

13:00 - 15:00

- **Frequency analysis**
- **Bode plot - 1st and 2nd order**
- Stability
- Stability margins

15:00 - 17:00

- Exercise **Day 6**

Day	Topics	Contents	Assignments
1	Introduction	Intro to feedback control, Pre-Test	MATLAB and Simulink
2	Control concepts	Regulators, block diagram, hand tuning	Hand tuning, Ziegler-Nichols
3	Transfer function and Laplace	Laplace and transfer function	Block diagrams and transfer functions
4	From time to frequency	Frequency domain properties	Introduction to REGBOT
5	Modelling	White/black box modelling and linearization	REGBOT – Transfer function from data
➡ **6**	**Frequency domain analysis**	**Bode plot, stability margins, P design**	**REGBOT – Bode, stability, P-Controller design**

What you will learn today

After this lecture, you will be able to:

- Interpret a **Bode** magnitude and phase plot
- Identify **poles and zeros** from Bode plots
- Construct asymptotic Bode plots
- Understand **stability** using gain margin and phase margin
- Relate **open-loop Bode plots to closed-loop stability**

Why Frequency Analysis?

The time-domain design is hard because:
Differential equations
High-order dynamics

The frequency domain allows:

Understanding **amplification vs frequency**

Robustness and noise sensitivity

Stability margins

How does the system react to **slow vs fast signals**?
What frequencies does the system **amplify or attenuate?**
How close is the system to **instability**?

Frequency analysis - *”real frequencies" in the Laplace domain*

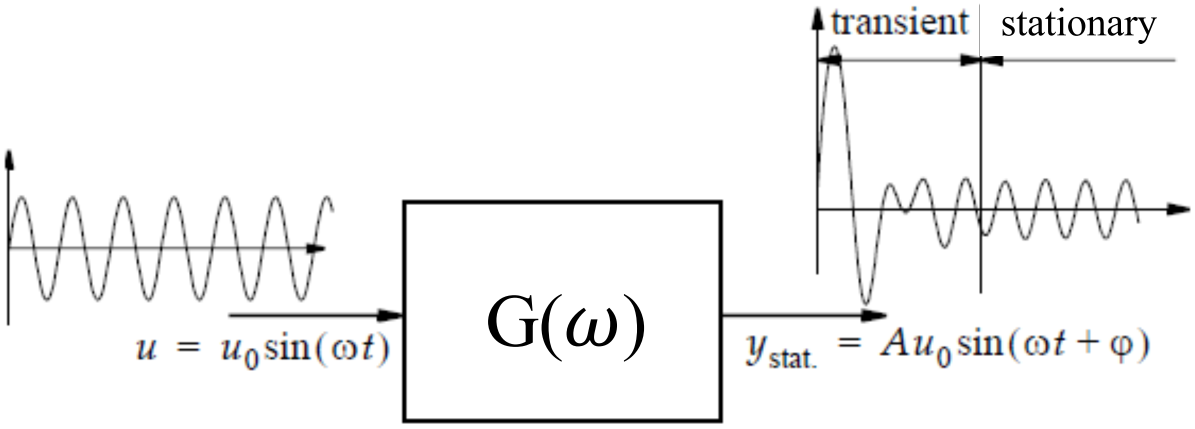
DTU

For sinusoidal input:

$$\mathbf{u} = \boldsymbol{u_0}\mathbf{sin}(\boldsymbol{\omega t})$$

The output is sinusoidal:

$$\mathbf{y} = |\mathbf{G}(\boldsymbol{\omega})|\, \boldsymbol{u_0}\mathbf{sin}(\boldsymbol{\omega}\mathbf{t} + \boldsymbol{\varphi})$$



Amplitude ratio (Magnitude):

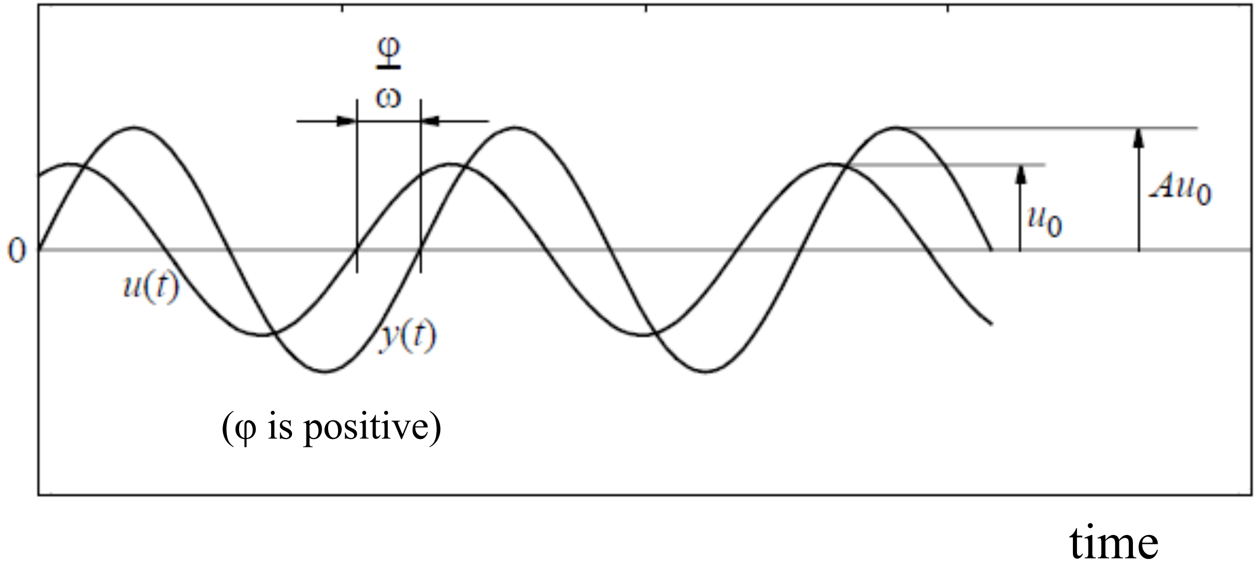
$$A = |G(\omega)|$$

Phase:

$$\varphi = argument\big(G(\omega)\big)$$

where:

$$G(\omega) = G(s)|_{s=j\omega}$$



A linear system's response to sinusoidal inputs - called the system's **frequency response** - can be obtained from knowledge of its pole and zero locations.

Poles and zeros in the s-plane

$$G(s) = \frac{s + 0.5}{(s+4)(s+1)}$$

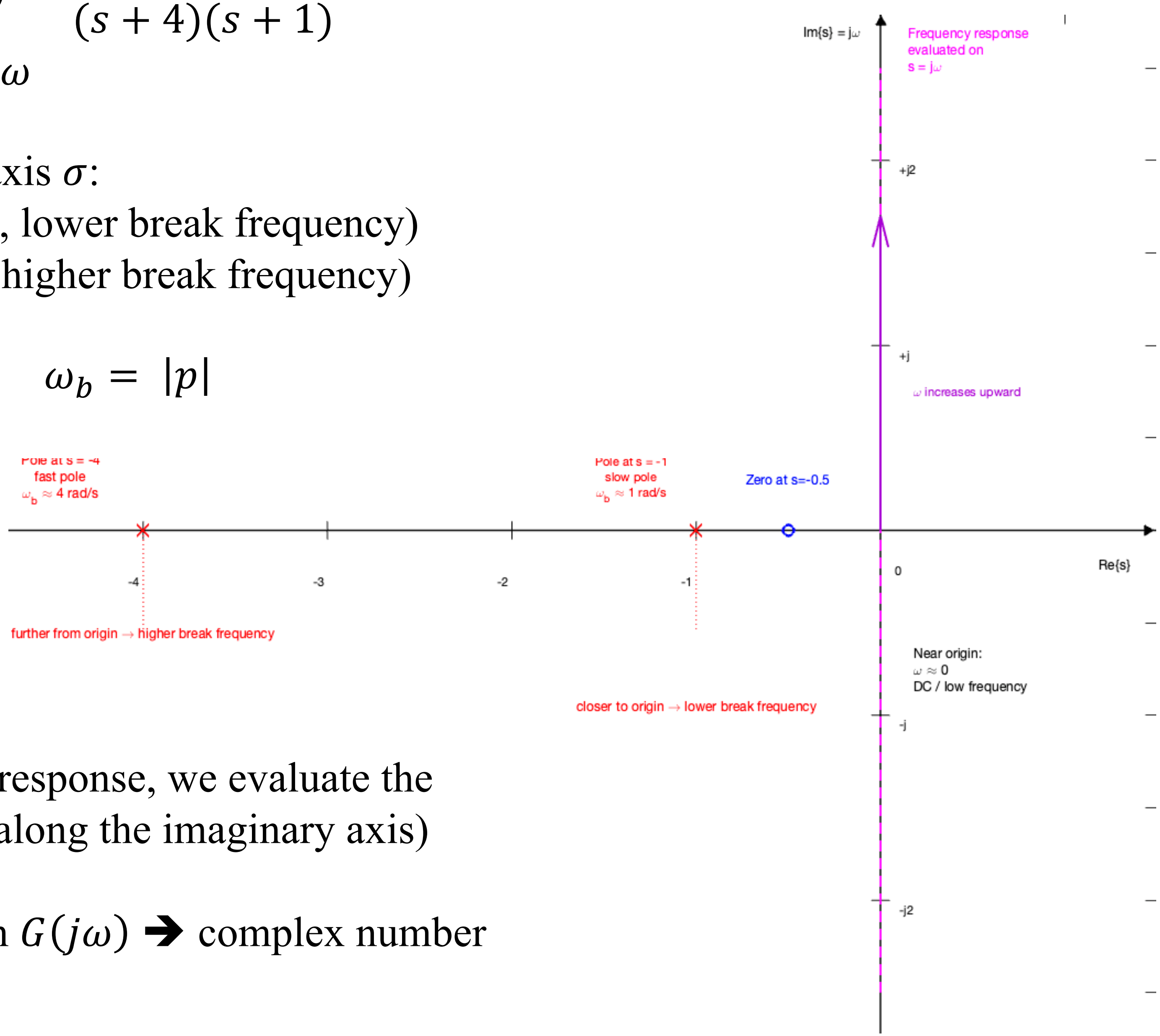
Complex variable $s = \sigma + j\omega$

Poles and zeros on the real axis σ:

P_1 ➔ s=-1 (slower dynamics, lower break frequency)

P_2 ➔ s=-4 (faster dynamics, higher break frequency)

$$\omega_b = |p|$$



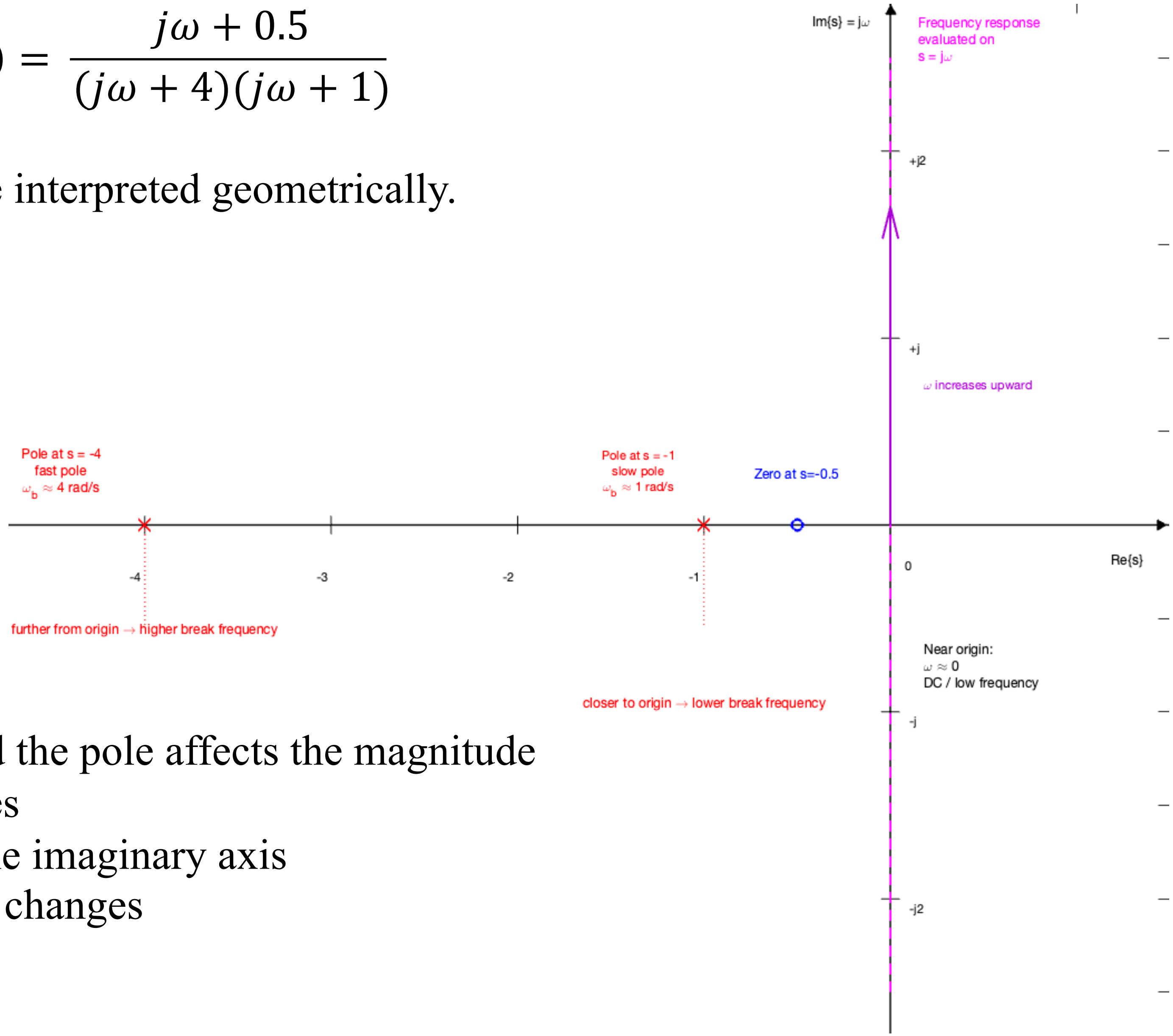
When computing frequency response, we evaluate the transfer function at $s = j\omega$ (along the imaginary axis)

So the Bode plot comes from $G(j\omega)$ ➔ complex number for every frequency ω

Poles and zeros in the s-plane

$$G(s) = \frac{j\omega + 0.5}{(j\omega + 4)(j\omega + 1)}$$

Frequency response can be interpreted geometrically.



For a pole at p:

$$|G(j\omega)| = \frac{1}{|j\omega + p|}$$

The distance between $j\omega$ and the pole affects the magnitude

When the frequency increases

- The point $j\omega$ moves up the imaginary axis
- The distance to each pole changes
- This changes the gain

The Bode plot is determined entirely by the location of poles and zeros in the s-plane

Process:

1. Start with the transfer function $G(s)$

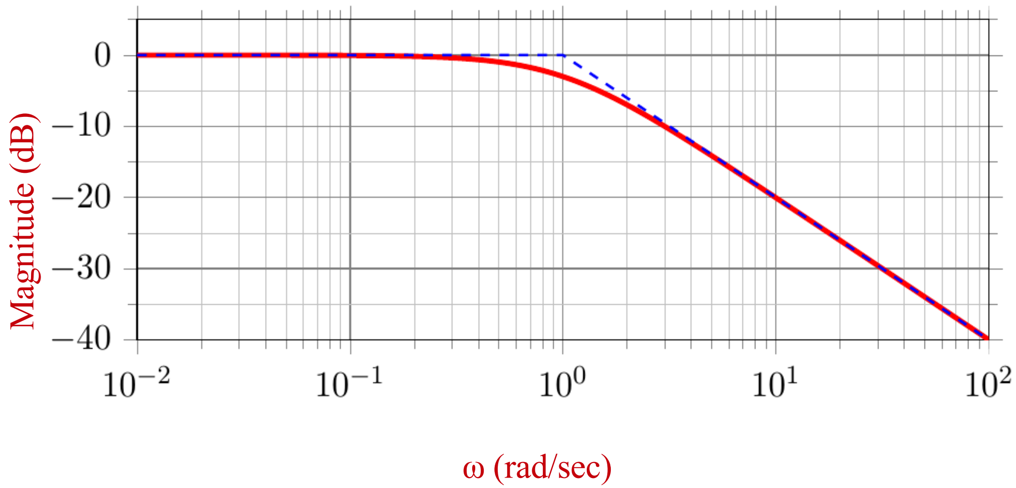
2. Identify poles and zeros

3. Evaluate on $s = j\omega$
4. Plot magnitude and phase vs frequency

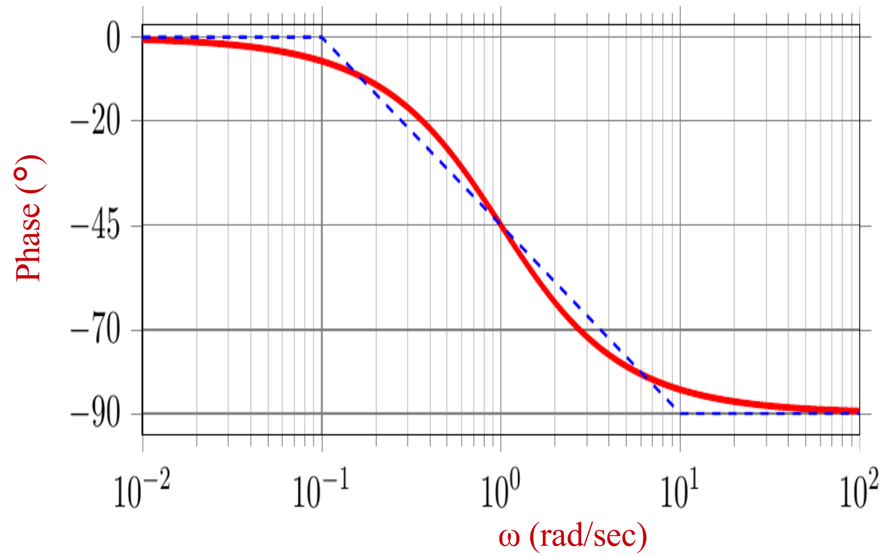
$$G(s) \rightarrow G(j\omega) \rightarrow Bode\ Plot$$

What is a Bode plot?

The magnitude curve is plotted in **logarithmic** scale



The phase curve is plotted in **linear** scale



The **Bode plot** shows the effect on the magnitude of the output compared to the input (gain) and the output's phase angle compared to the input’s phase

These values are presented as a function of the frequency ω $\left(\frac{rad}{sec}\right)$

The magnitude is in dB, and the phase is in degrees

Why the Logarithmic Scale?

Wide Range: it covers frequencies from 0.01 to 10,000 rad/s on one page

Decades: Each equal horizontal distance is a factor of 10

Straight Lines: Makes complex transfer functions look like simple linear segments (asymptotes)

Bode plot – Pole effect

Below w_b *the system acts like a constant gain*
Above w_b *the* frequency effect dominates

$$G(s) = \frac{1}{s + \omega_0}$$

$$G(s) = \frac{1}{s + 1}$$

$$G(j\omega) = \frac{1}{j\omega + 1}$$

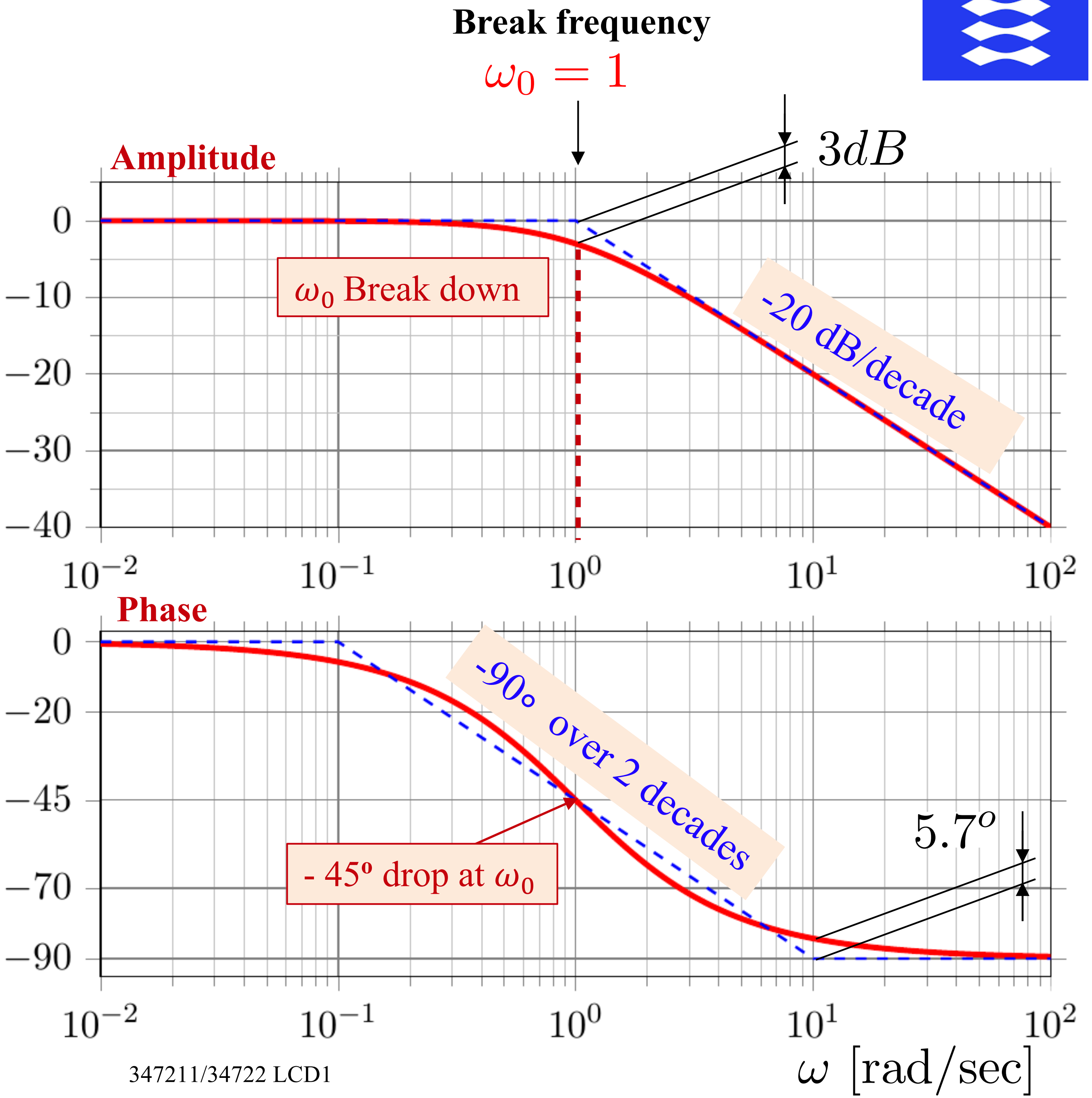
$$G(j1) = \frac{1}{j + 1}$$

$$G(j1) = \frac{1}{2}(1 - j)$$

$$G(j1) = \frac{\sqrt{2}}{2} e^{-\frac{\pi}{4}}$$

$$G(j1) = 0.707\angle - 45^o$$

$$|G(j1)|_{dB} = -3dB$$



Bode plot – Zero effect

$$G(s) = s + \omega_0$$

$$G(s) = s + 0.2$$

$$G(j\omega) = j\omega + 0.2$$

$$G(0.2j) = 0.2j + 0.2$$

$$G(0.2j) = 0.283e^{\frac{\pi}{4}}$$

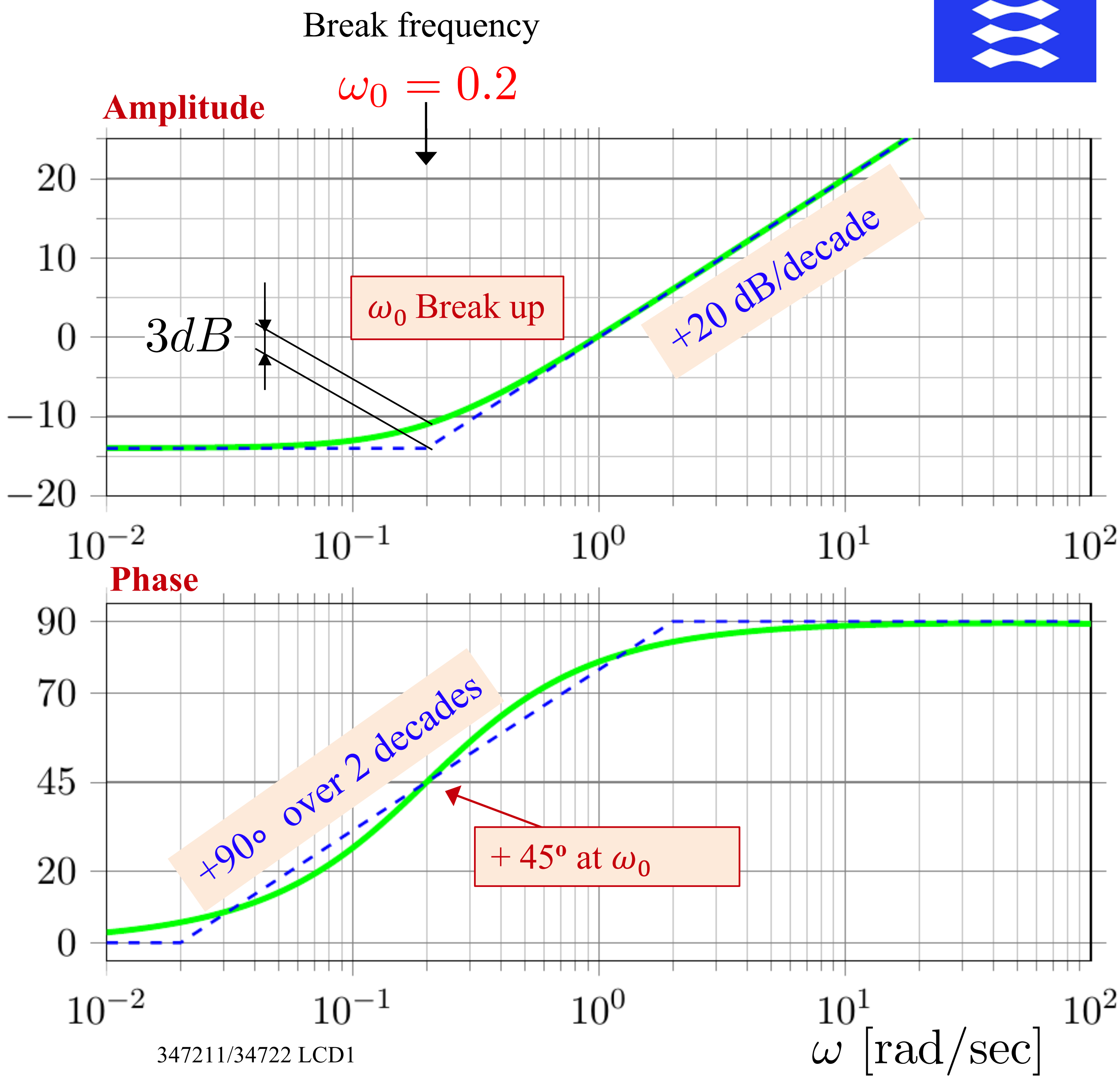
$$G(0.2j) = 0.283\angle 45^o$$

$$|G(0.2j)|_{dB} = -10.97dB$$

MATLAB

A = abs(0.2*j + 0.2)

F = angle(0.2*j + 0.2)



Bode plot – Combining pole and zero

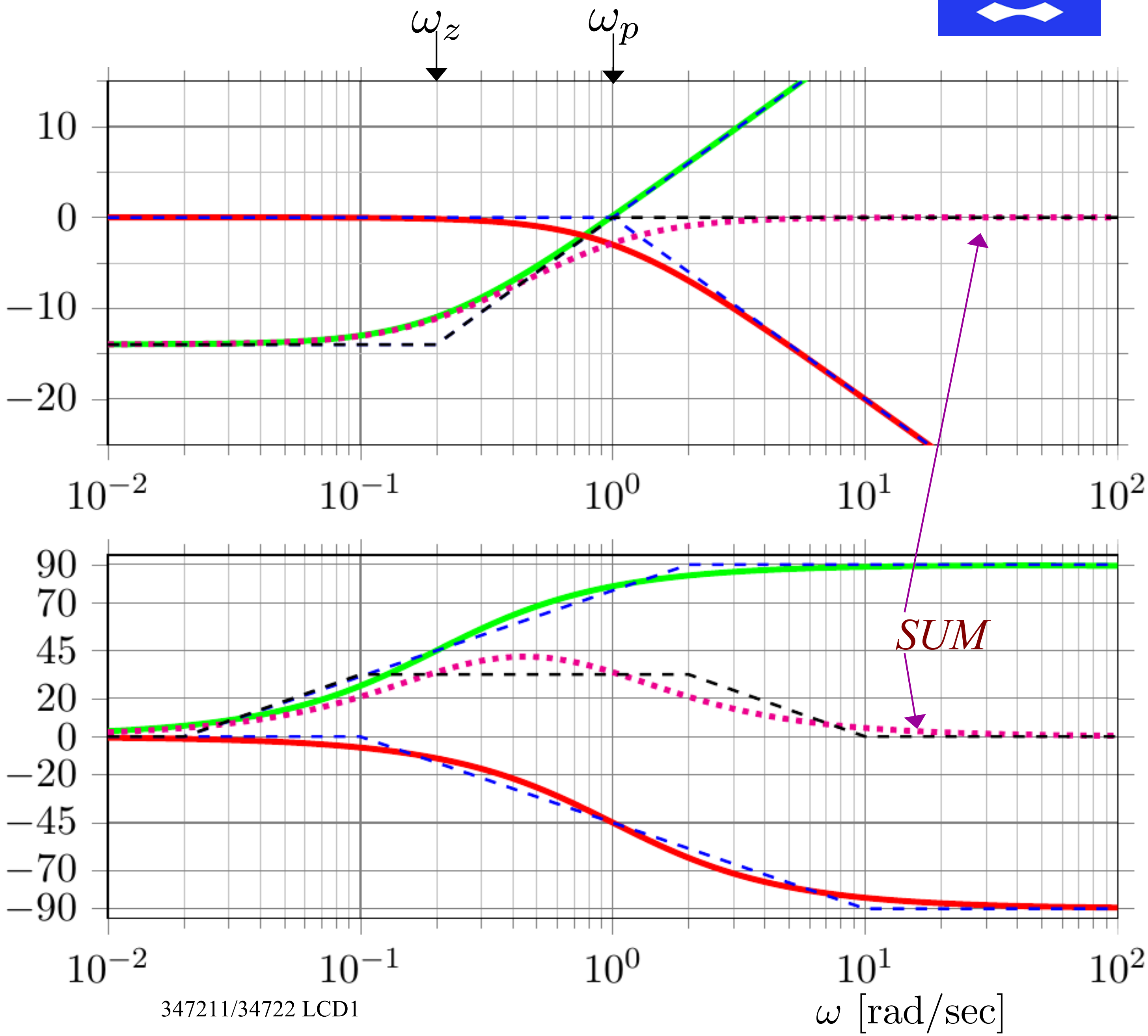
The product of transfer functions becomes a summation of curves

$$G(s) = \frac{s+\omega_z}{s+\omega_p}$$

$$\rightarrow \boxed{G(s)} \rightarrow$$

$$\rightarrow \boxed{\frac{1}{s+1}} \rightarrow \boxed{s+0.2} \rightarrow$$

$$G(s) = \frac{s+0.2}{s+1}$$



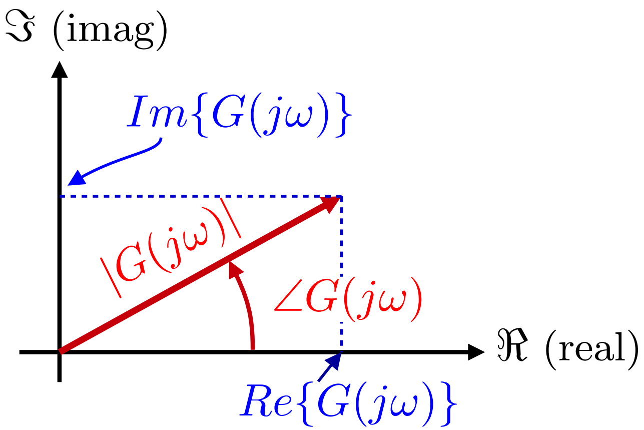
MATLAB

```
figure(1)
Gp = tf([1],[1 1])
Gz = tf([1 0.2],[1])
G = Gp * Gz
bode(Gp, Gz, G)
```

Magnitude (Amplitude)

$$|G(j\omega)| = \sqrt{Re^2 + Im^2}$$

$$|G(j\omega)|_{dB} = \boxed{20\log_{10}(|G(j\omega)|)}$$



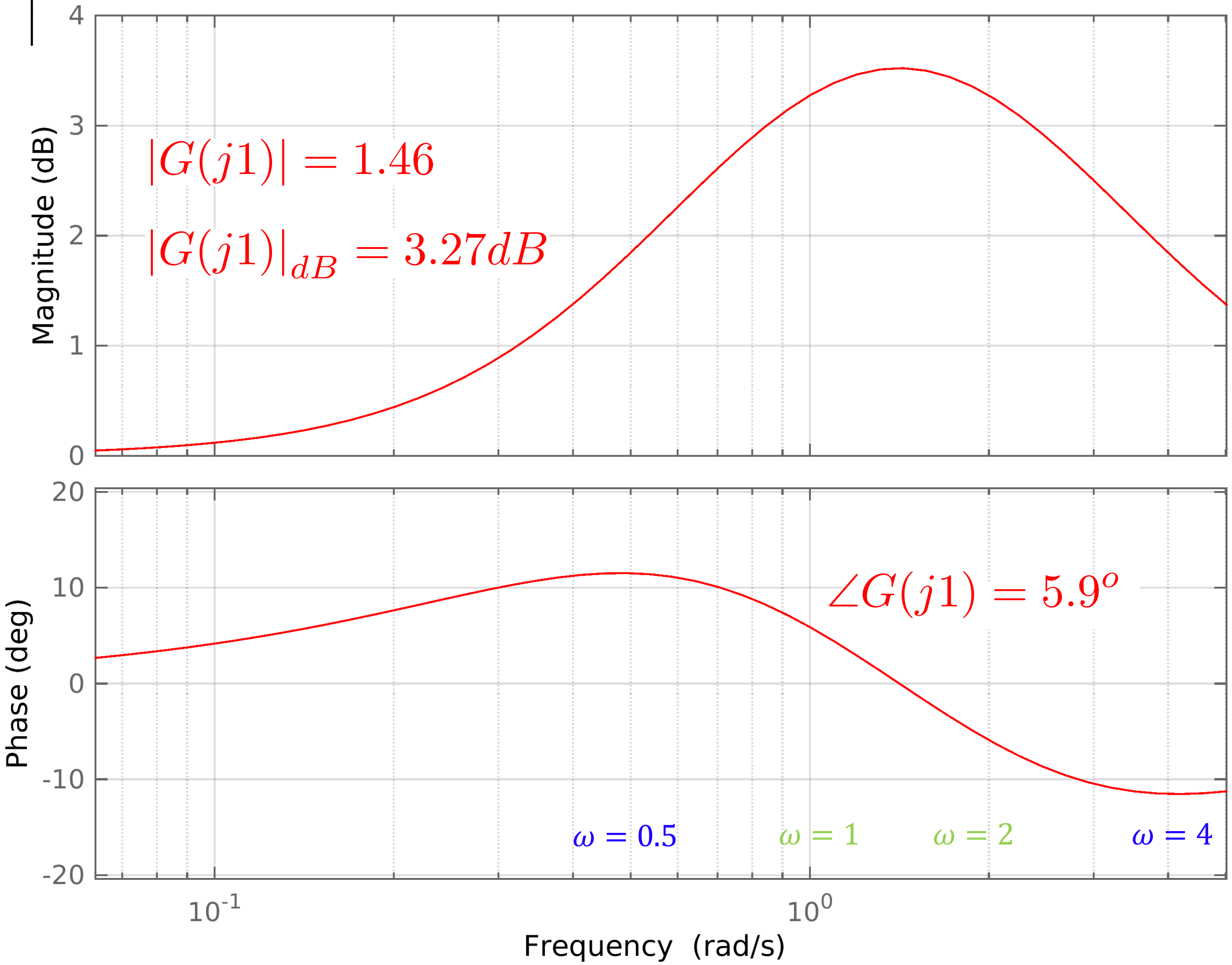
Phase

$$\angle G(j\omega) = \boxed{\tan^{-1}\left(\frac{Im\{G(j\omega)\}}{Re\{G(j\omega)\}}\right)}$$

Bode plot (2 poles and 2 zeros on the LHP)

$$|G(j\omega)| = \left| \frac{(j\omega+0.5)(j\omega+4)}{(j\omega+1)(j\omega+2)} \right|$$

$$G(j1) = 1.46e^{0.103j}$$



Bode plot - zero point in the right half-plane

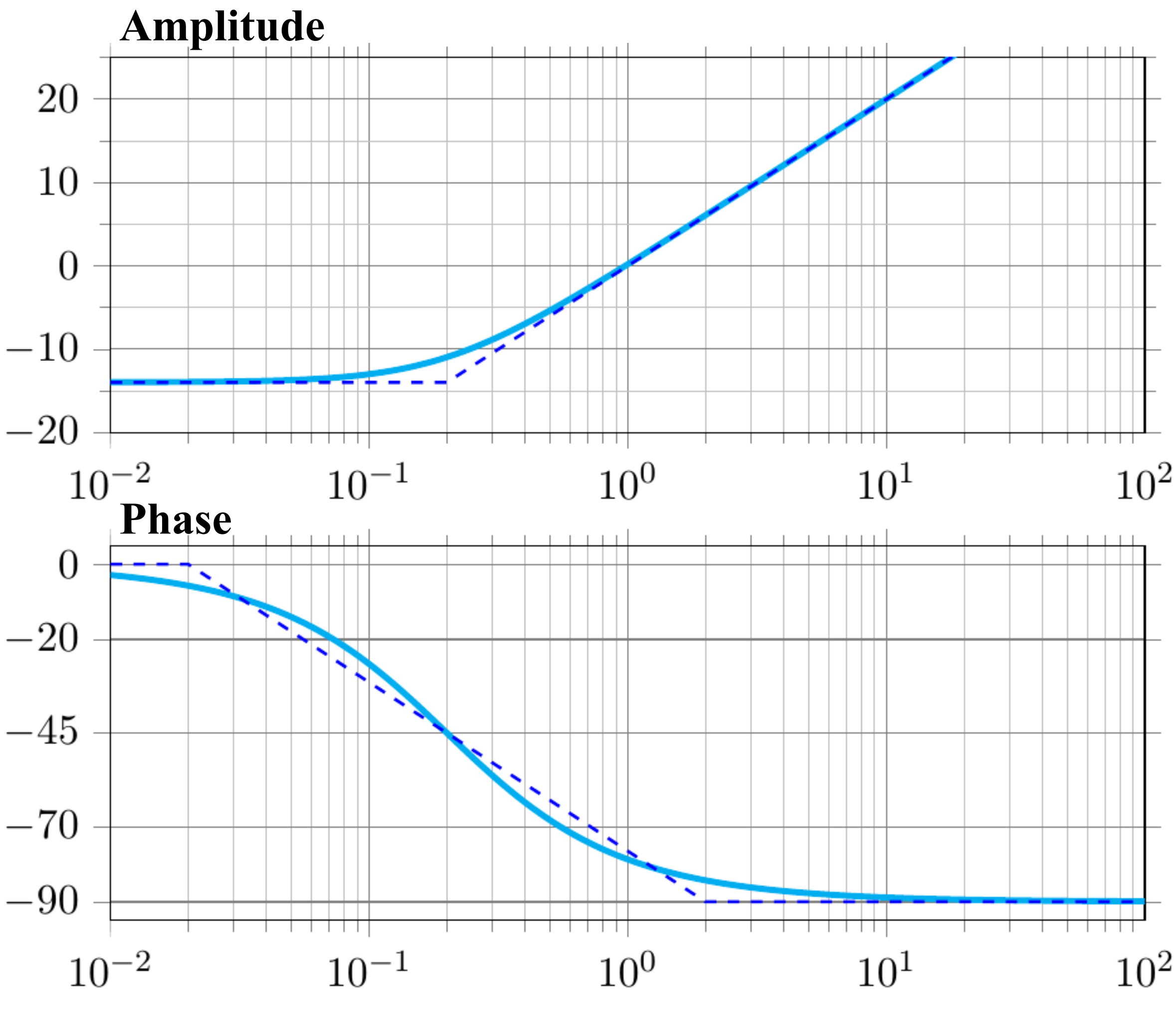
$$G(s) = -s + \omega_0$$

$$G(s) = -s + 0.2$$

$$G(j\omega) = -j\omega + 0.2$$

Zero point in the right half-plane:

Amplitude unchanged, however
Phase goes **90º DOWN**!



Pole in the right half-plane?

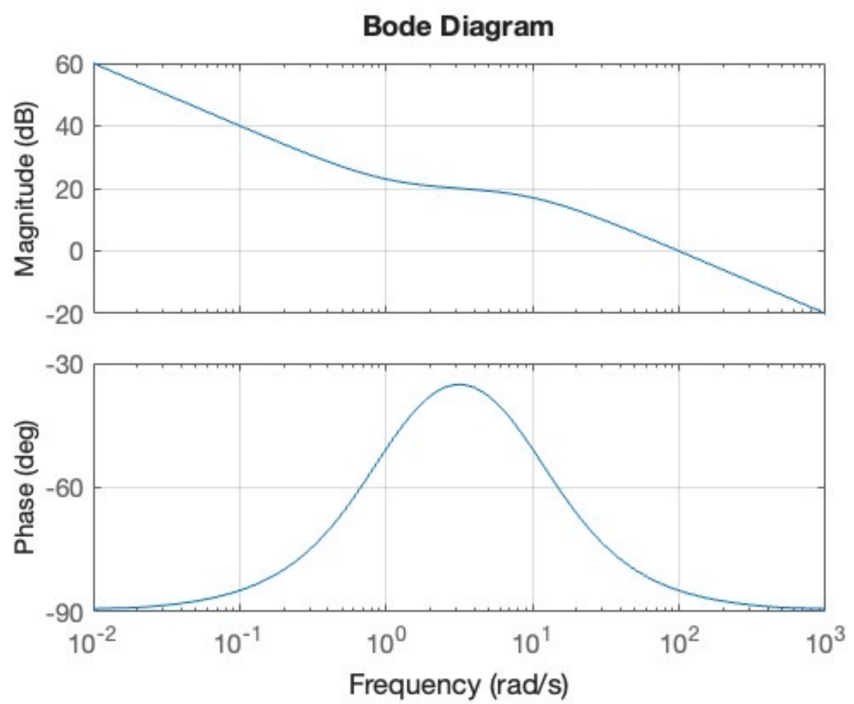
The magnitude goes down and the phase up

$$G = \frac{100s + 100}{s^2 + 10s} = \frac{10\,(s+1)}{s(1+0.1s)}$$

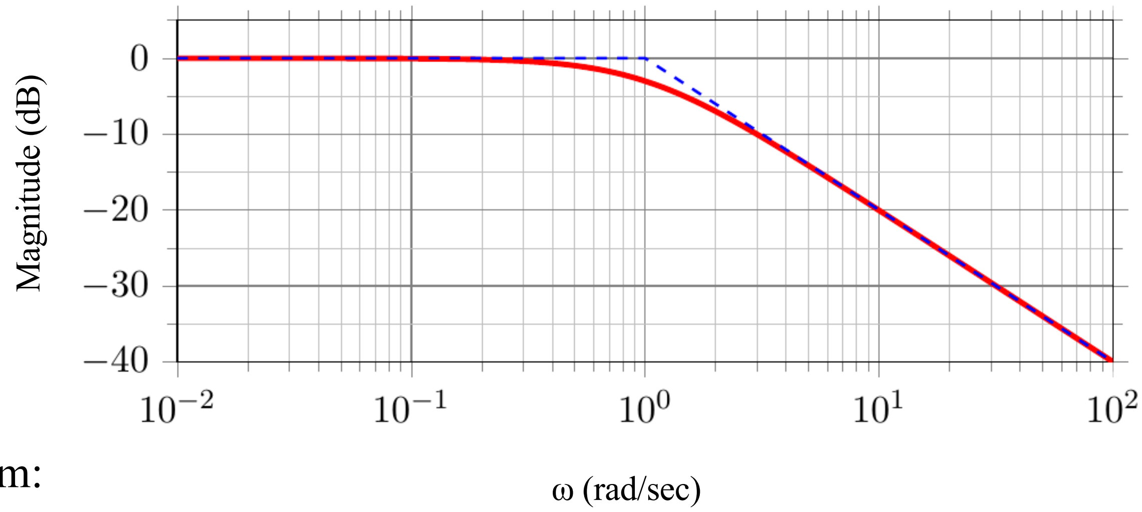
Element	**Location**	**Effect**
Integrator	s= 0	−20 dB/dec
Zero	s= −1	+20 dB/dec
Pole	s= −10	−20 dB/dec

start: **−20 dB/dec**
after $\boldsymbol{\omega = 1}$: slope becomes **0**
after $\boldsymbol{\omega = 10}$: slope becomes **−20 dB/dec**

Watch the slope. The integrator drops the magnitude. The zero at $\boldsymbol{\omega = 1}$ cancels that downward slope (making it flat). Then the pole at $\boldsymbol{\omega = 10}$ drops the magnitude again.



Bode plot – Magnitude curve Cookbook



The open-loop transfer function in the Bode form:

$$G(j\omega) = K_0 (j\omega)^n \frac{(j\omega\tau_1 + 1)\ldots}{(j\omega\tau_a + 1)\ldots}$$

K_0 is the gain at ω=0 and is equal to DC gain

1. Mark the break frequencies corresponding to poles and zeros on the ω axis
2. Determine the initial slope
   - ***n*** *= # of zeros at origin - # poles at origin*
   - If **n=0,** the magnitude starts as a horizontal line at **$20 \log_{10} |K_0|$**
   - If $\boldsymbol{n \neq 0}$, the magnitude starts with a slope **20n dB/dec** passing through **$20 \log_{10} |K_0|$ at ω=1**
3. Modify the slope at each break frequency

   pole ➔ -20 dB/dec

   zero ➔ +20 dB/dec

Bode plot - Phase curve - Cookbook

1. Mark the break frequencies of poles and zeros on the ω axis

2. Determine the initial phase at low frequency:

- **n = # of zeros at origin - # poles at origin**
- If $K_0 > 0$: $\phi = -90n^{\circ}$
- If $K_0 < 0$: $\phi = -90n^{\circ}-180^{\circ}$

The phase remains approx. constant until the first break frequency

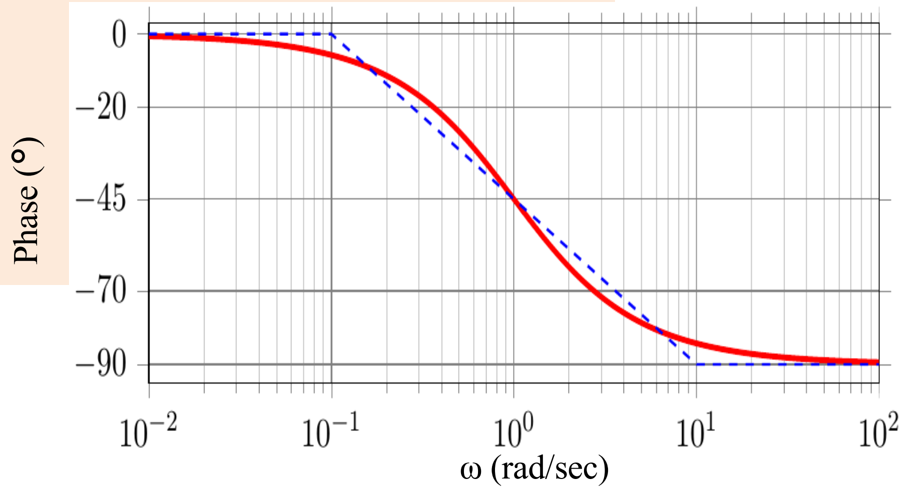
3. Modify the phase at each break frequency

Pole (LHP) ➔ -90°

Zero (LHP) ➔ +90°

Pole (RHP) ➔ +90°

Zero (RHP) ➔ -90°



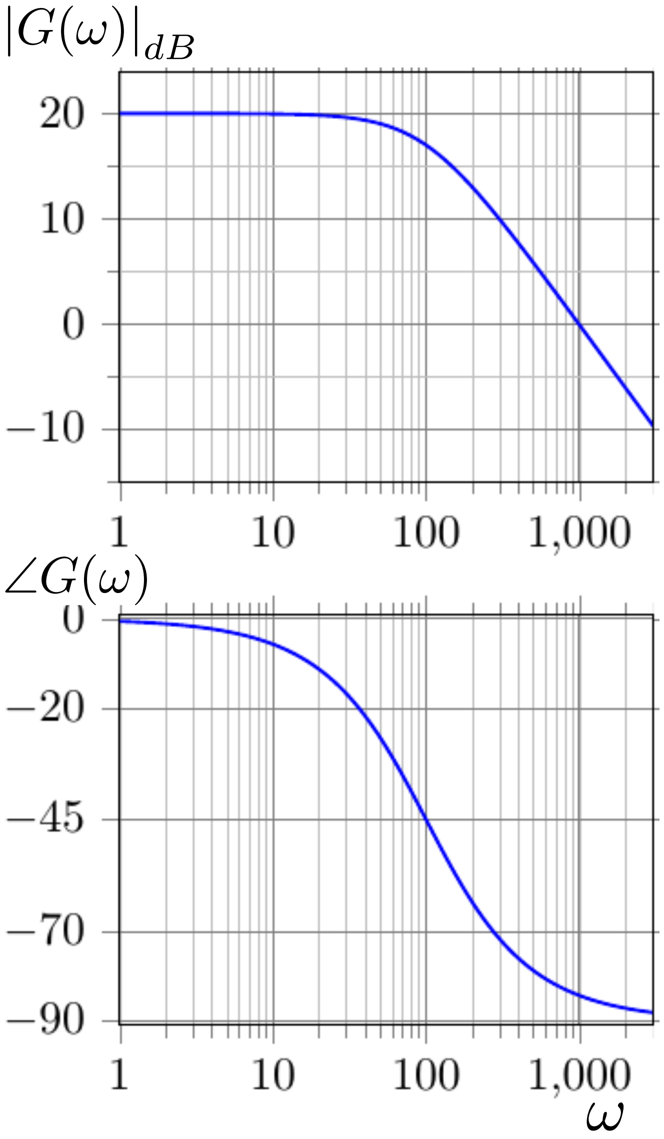
Singularity	Magnitude	Phase
Zero left	**↗ (+20 dB/dec)**	**↗ (+90°)**
Pole left	**↘ (-20 dB/dec)**	**↘ (- 90°)**
Zero right	**↗ (+20 dB/dec)**	**↘ (- 90°)**
Pole right	**↘ (-20 dB/dec)**	**↗ (+90°)**

Effect of poles and zeros on asymptotic Bode plot of magnitude and phase

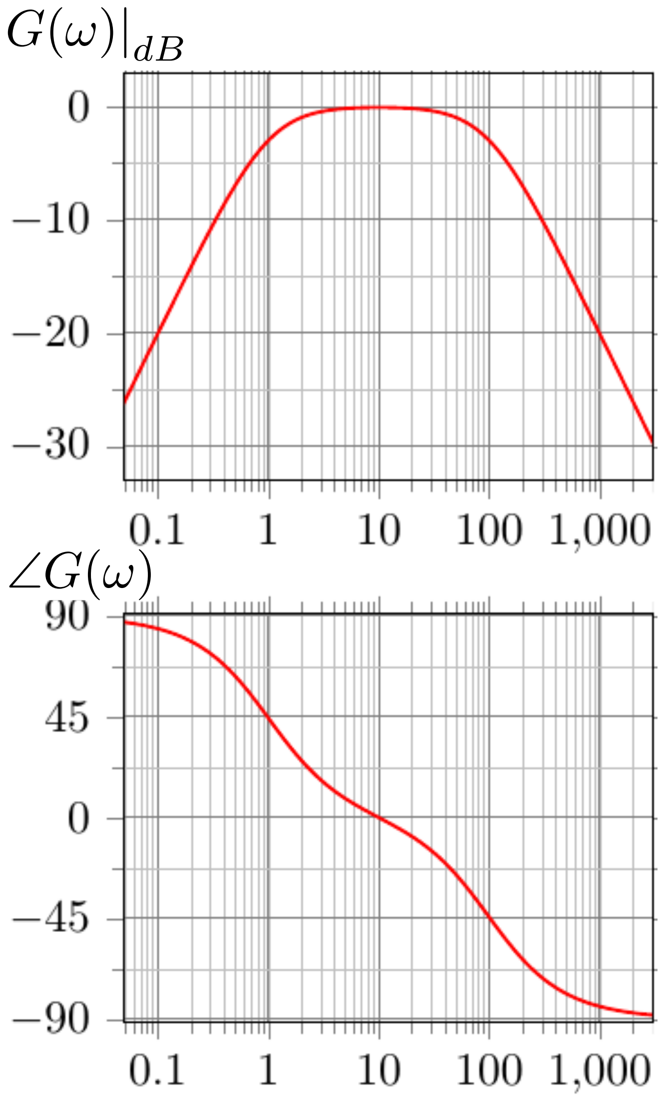
Singularity	Magnitude	Phase
Gain K	**0 dB/dec**	**0°**

Reading the transfer functions from Bode
Identify Initial Slope: it determines the number of integrators/differentiators.
Identify Final Slope: it determines the relative degree.
Identify Corner Frequencies: locate the poles and zeros.

Control Question – Find the Transfer Function



$G(s) = ?$



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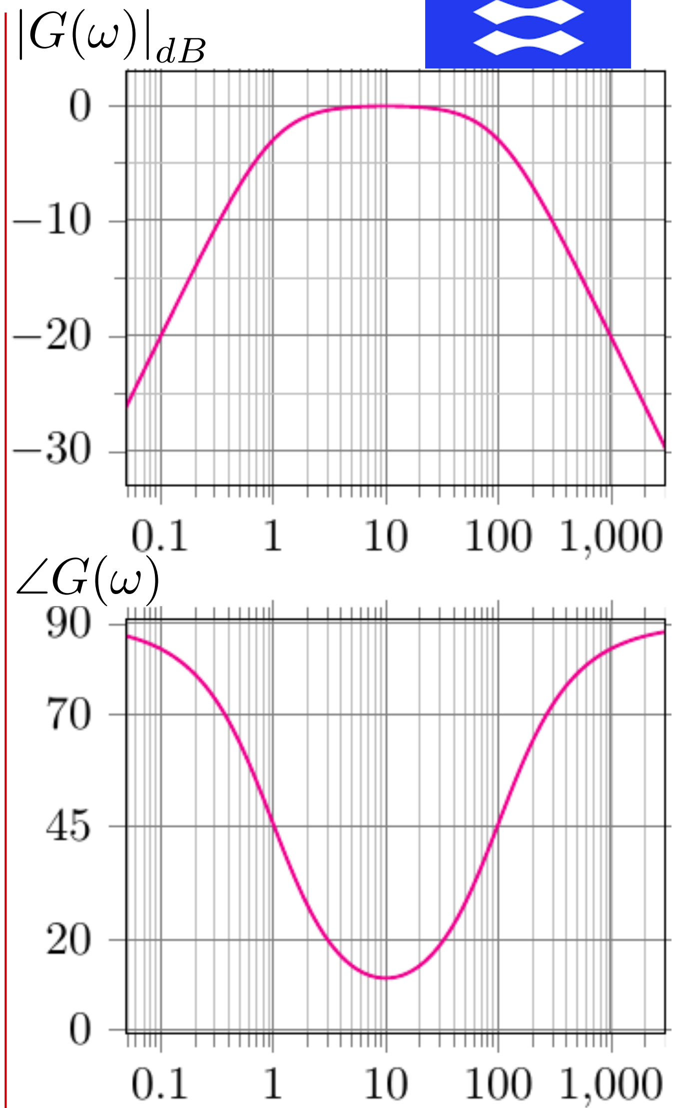
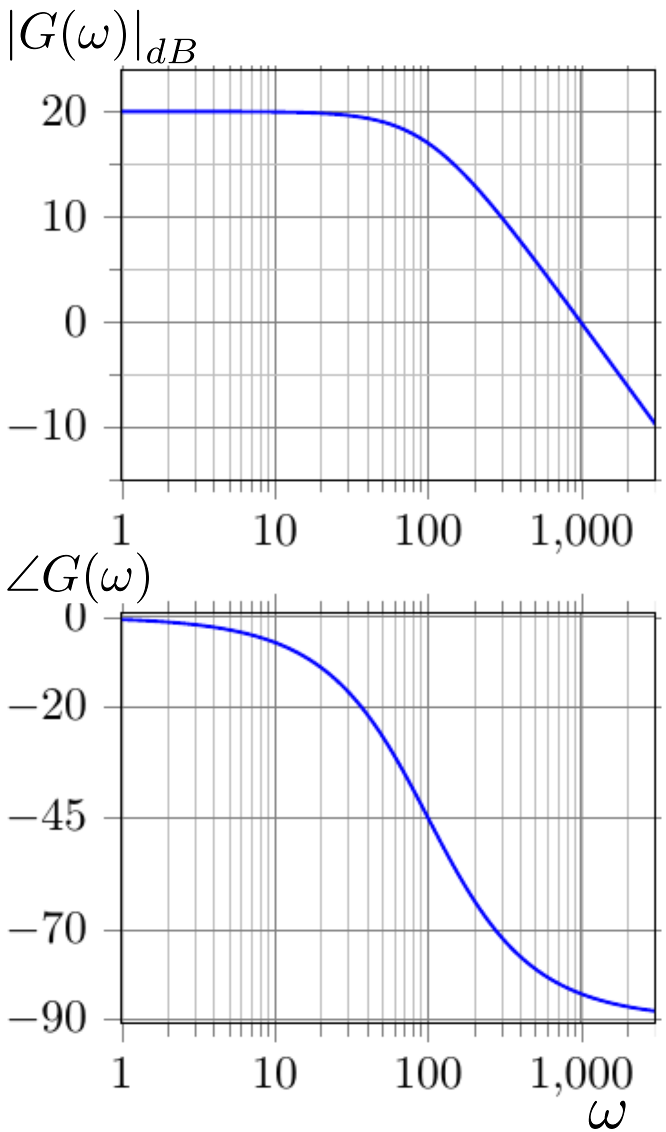


$G(s) = ?$

Control Question – Find The Transfer Function



Break down 20dB / decade and phase down 90 degrees, fits with a pole.

Break frequency at 3dB change = 100 rad / sec

$$G(s) = \frac{k}{s+100}$$

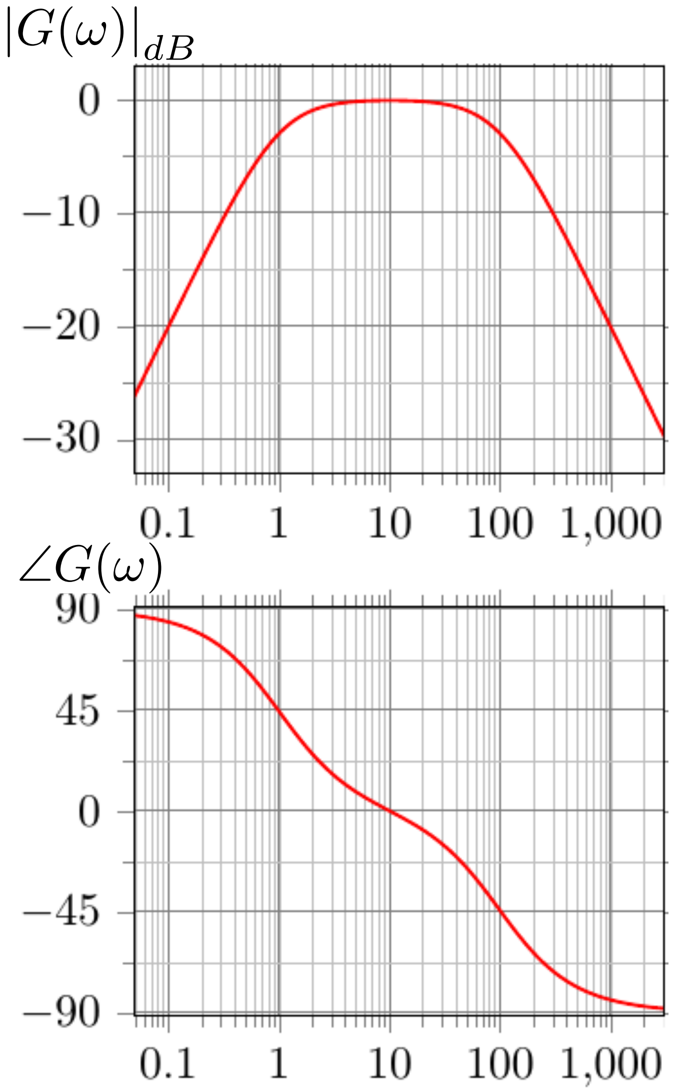
$$s = j\omega \quad \text{and} \qquad \lim_{\omega \to 0} G(\omega) = 20 \text{ dB}$$

$$\lim_{s \to 0} G(s) = \frac{k}{100} = 10^{\frac{20}{20}} \Rightarrow K = 1000$$

$$10^{\frac{M_{db}}{20}}$$

$$G(s) = \frac{1000}{s+100}$$

Control Question - Find The Transfer Function



Break down 20 dB/dec and phase down by 90° degrees 1 and 100 rad/sec → **2 poles at s = -1 and s = -100**

It starts with +20 dB/dec, must start with a bend up, fits with a starting angle of +90 degrees → **zero point in s = 0**

$$G(s) = \frac{k(s+0)}{(s+1)(s+100)}$$

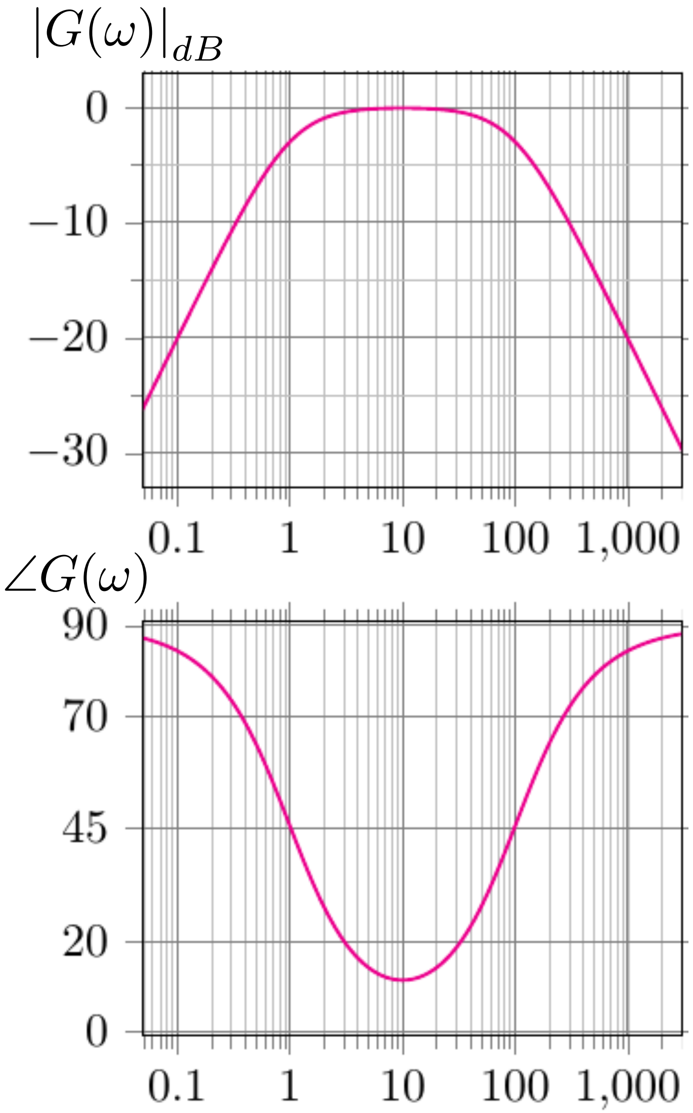
$$\omega = 10 \qquad |G(\omega)| = 0dB = 1$$

$$\left| \frac{ks}{(s+1)(s+100)} \right|_{s=10j} = 1$$

$$\left| \frac{k(10j)}{(10j)(100)} \right| = 1 \Rightarrow k = 100$$

$$G(s) = \frac{100s}{(s+1)(s+100)}$$

Control Question - Find The Transfer Function



In terms of amplitude, the curve is similar to the previous one.
The phase also fits, except for the last pole.

The last pole must be in the right half-plane!

$$G(s) = \frac{100s}{(s+1)(s-100)}$$

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Lesson 6/13 – Bode Plot and P-controller Design

Schedule

13:00 - 15:00

- Frequency analysis
- **Bode plot - 1st and 2nd order**
- Stability
- Stability margins

15:00 - 17:00

- Exercise **Day 6**

Complex poles

S-plane
(complex frequency plane)

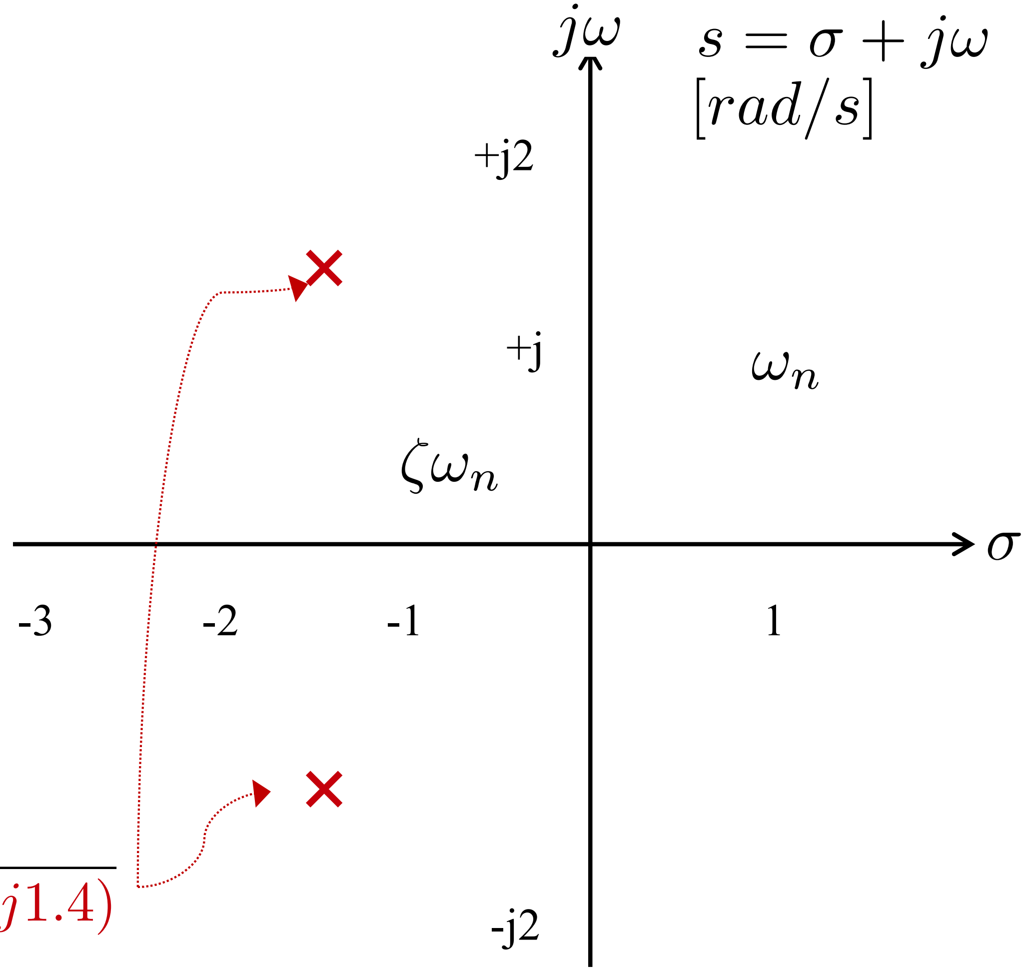
$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\zeta = 0.707$$

$$\omega_n = 2 \ [rad/s]$$

$$G(s) = \frac{4}{(s^2 + 2 \cdot 0.707 \cdot 2s + 4)}$$

$$G(s) = \frac{4}{(s + 1.4 + j1.4)(s + 1.4 - j1.4)}$$



(2nd order poles always come in complex conjugated pairs)

S-plane - plot 1

$$G = \frac{s+0.5}{(s+1)^2}$$

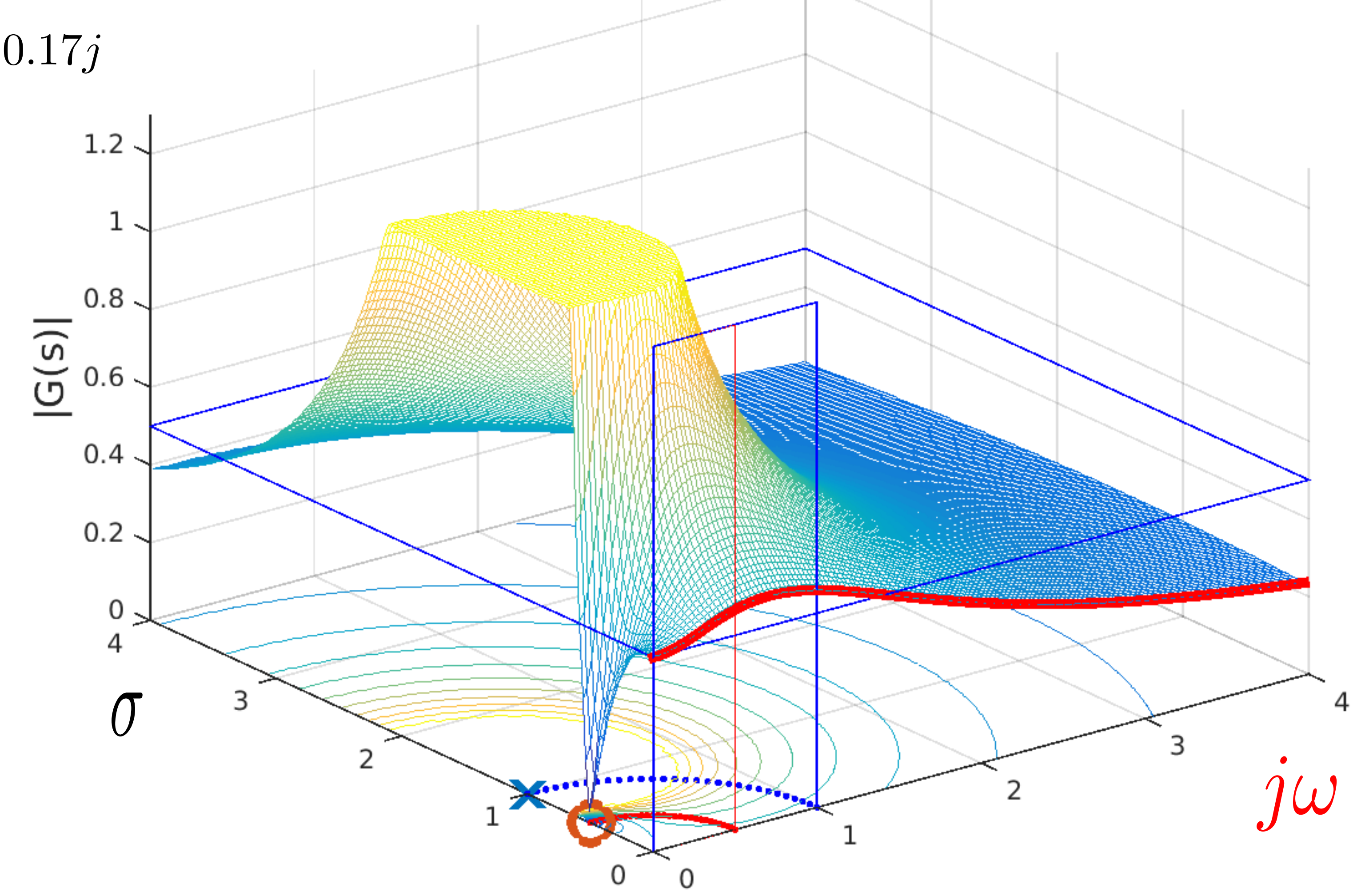
$$G = \frac{s+0.5}{s^2+2s+1}$$

$\omega_n = 1$

$P = -0.98 \pm 0.17j$

$Z = -0.5j$

$\zeta = 1$



S-plane - plot 2

$$G = \frac{s+0.5}{(s+0.98+0.17j)(s+0.98-0.17j)}$$

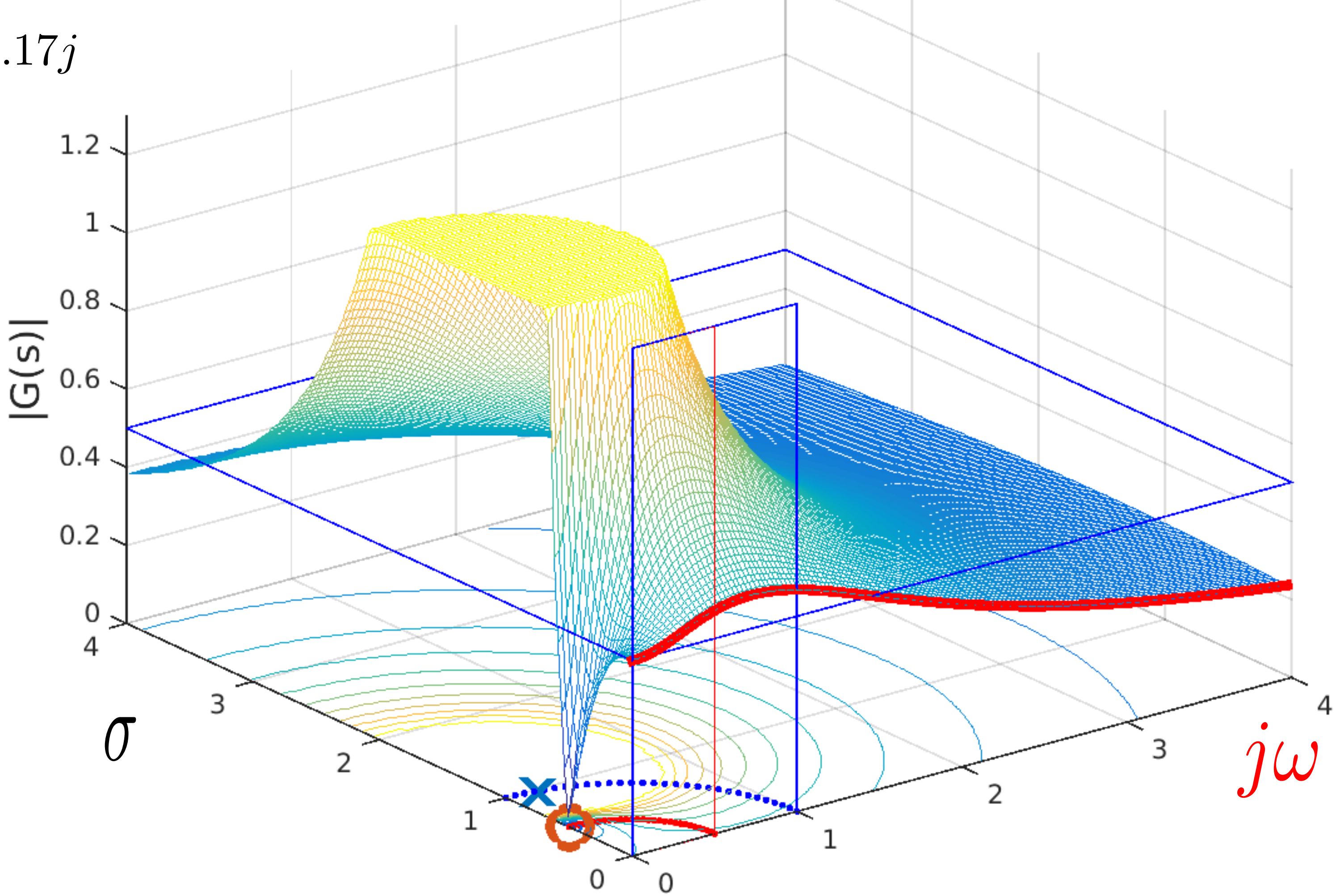
$$G = \frac{s+0.5}{s^2+2\cdot 0.98s+1}$$

$\omega_n = 1$

$P = -0.98 \pm 0.17j$

$Z = -0.5j$

$\zeta = 0.98$



S-plane - plot 3

$$G = \frac{s+0.5}{(s+0.94+0.34j)(s+0.94-0.34j)}$$

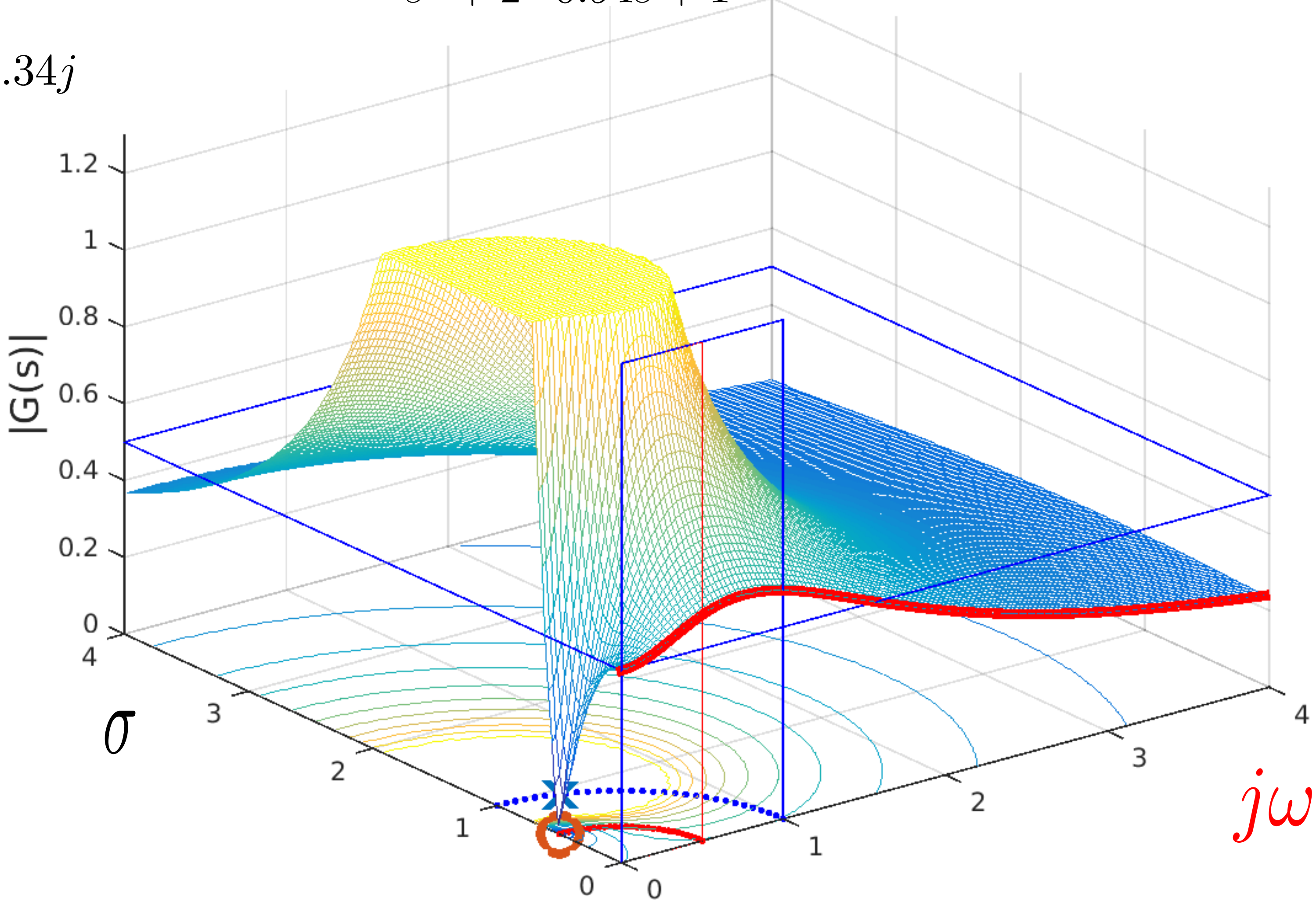
$$G = \frac{s+0.5}{s^2+2\cdot 0.94s+1}$$

$\omega_n = 1$

$P = -0.94 \pm 0.34j$

$Z = -0.5j$

$\zeta = 0.94$



S-plane - plot 4

$$G = \frac{s + 0.5}{(s + 0.87 + 0.5j)(s + 0.87 - 0.5)}$$

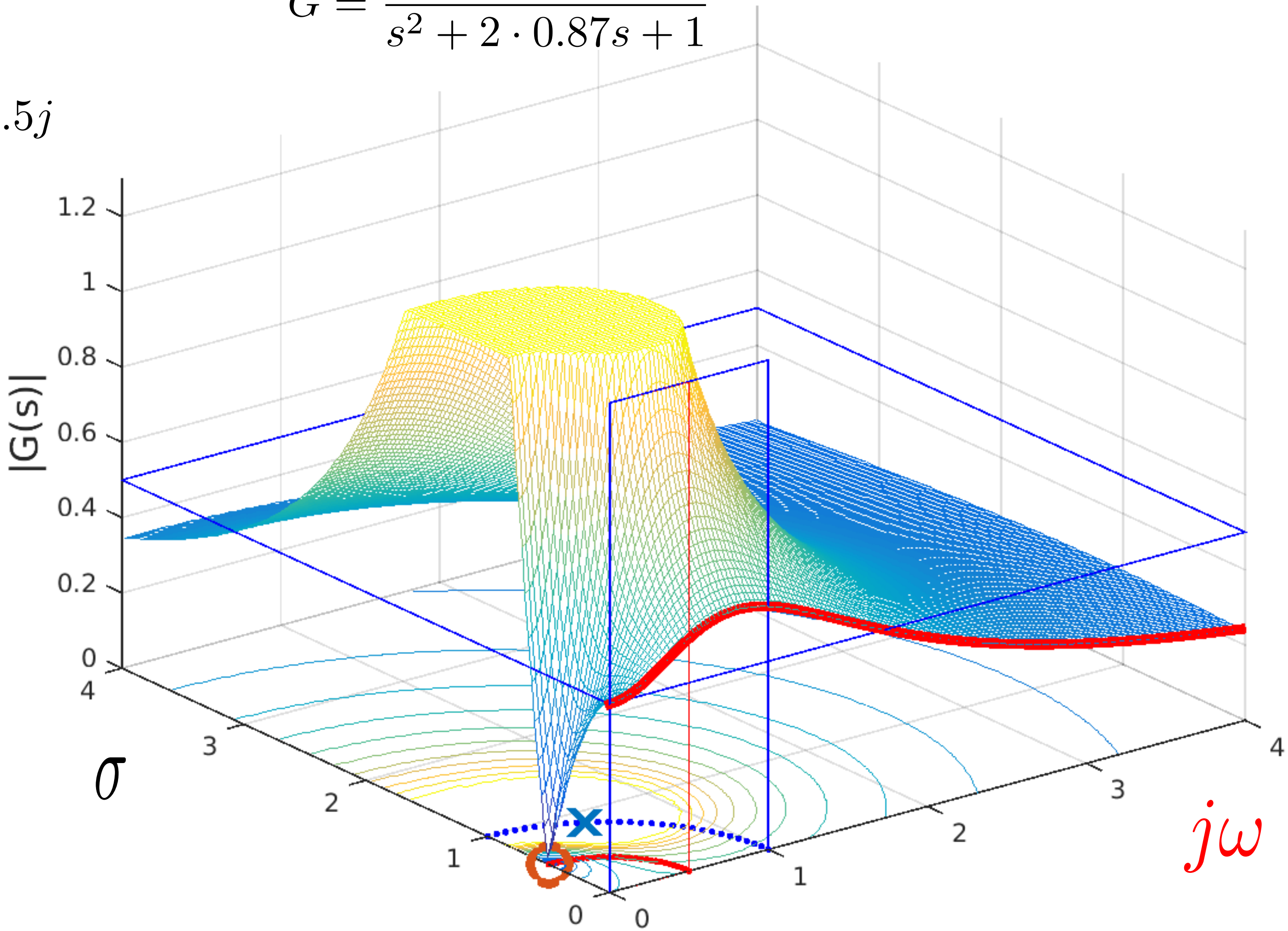
$$G = \frac{s + 0.5}{s^2 + 2 \cdot 0.87s + 1}$$

$\omega_n = 1$

$P = -0.87 \pm 0.5j$

$Z = -0.5j$

$\zeta = 0.87$



S-plane - plot 5

$$G = \frac{s+0.5}{(s+0.77+0.64j)(s+0.77-0.64)}$$

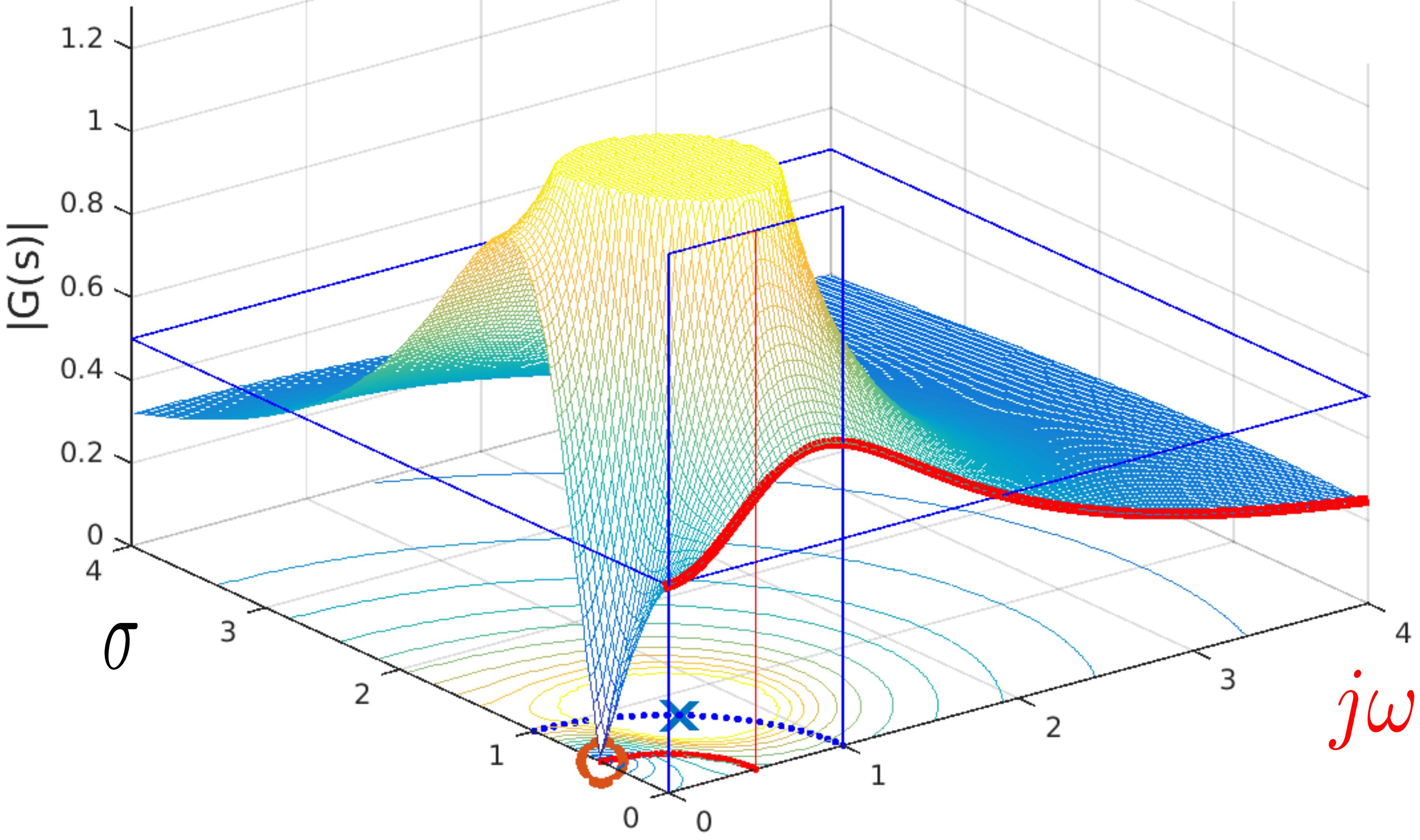
$$G = \frac{s+0.5}{s^2+2\cdot 0.77s+1}$$

$\omega_n = 1$

$P = -0.77 \pm 0.64j$

$Z = -0.5j$

$\zeta = 0.77$



S-plane - plot 6

$$G = \frac{s+0.5}{(s+0.64+0.77j)(s+0.64-0.77)}$$

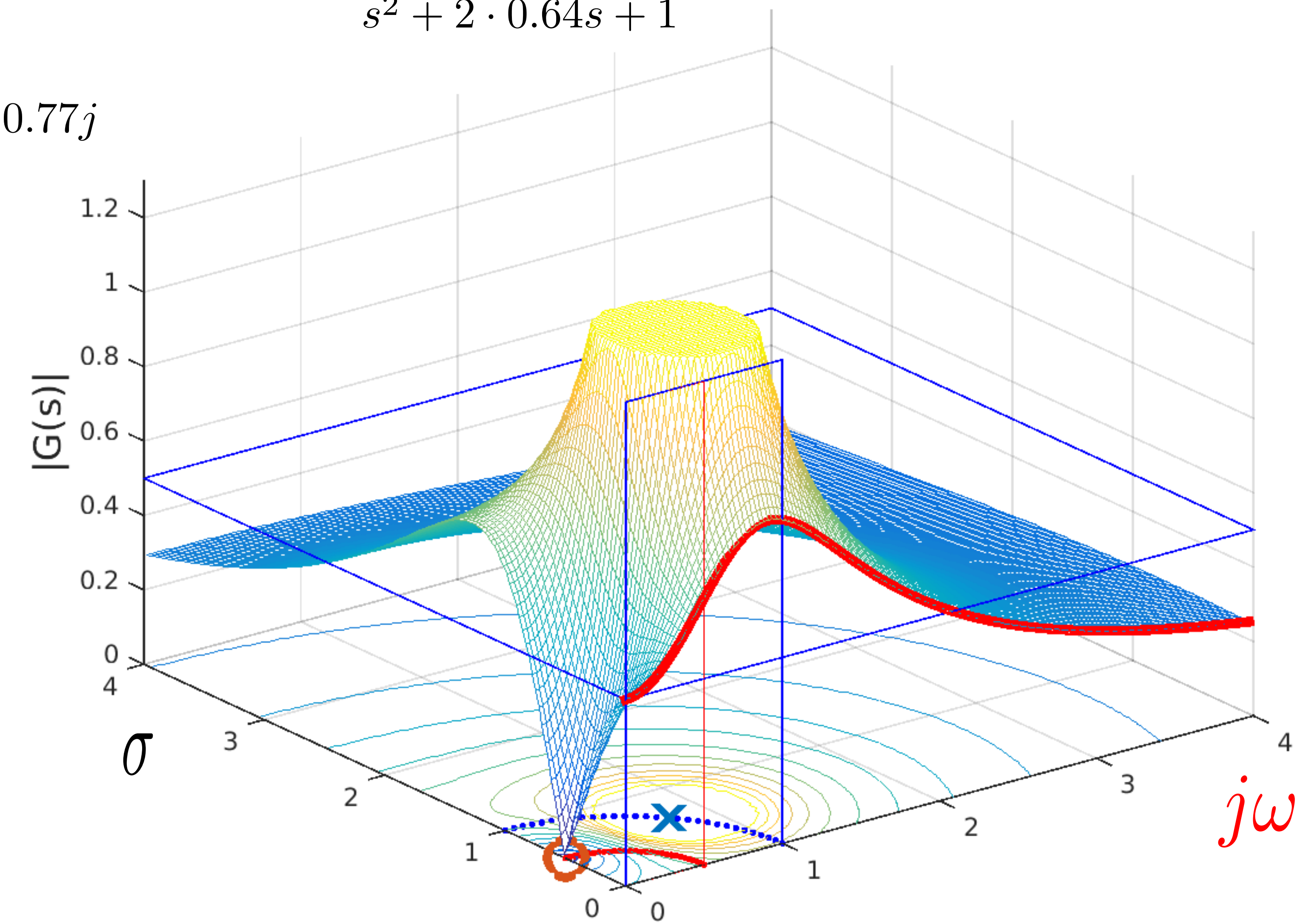
$$G = \frac{s+0.5}{s^2+2\cdot 0.64s+1}$$

$\omega_n = 1$

$P = -0.64 \pm 0.77j$

$Z = -0.5j$

$\zeta = 0.64$



S-plane - plot 7

$$G = \frac{s + 0.5}{(s + 0.5 + 0.87j)(s + 0.5 - 0.87j)}$$

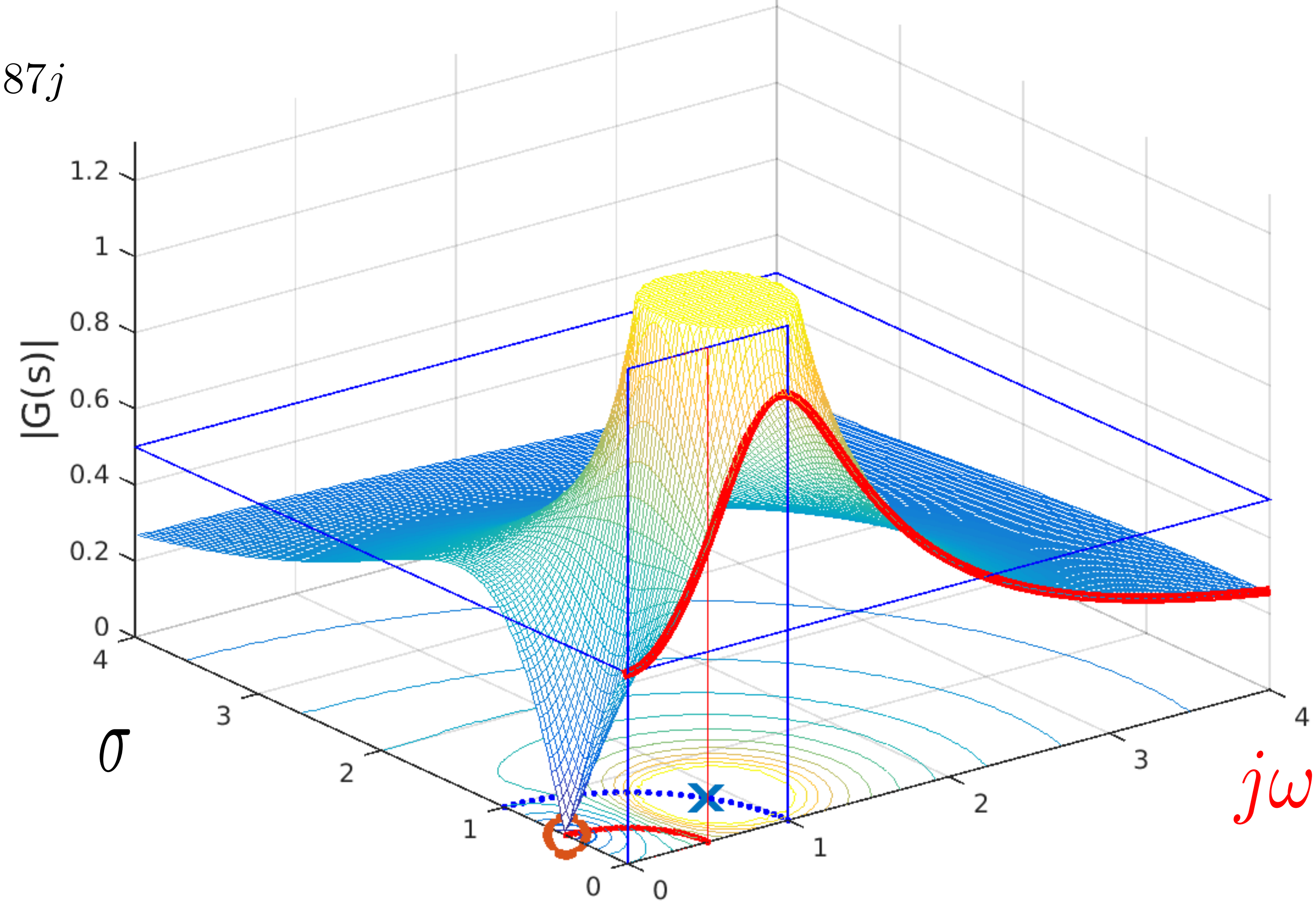
$$G = \frac{s + 0.5}{s^2 + 2 \cdot 0.5 s + 1}$$

$\omega_n = 1$

$P = -0.5 \pm 0.87j$

$Z = -0.5j$

$\zeta = 0.5$



S-plane - plot 8

$$G = \frac{s + 0.5}{(s + 0.34 + 0.94j)(s + 0.34 - 0.94j)}$$

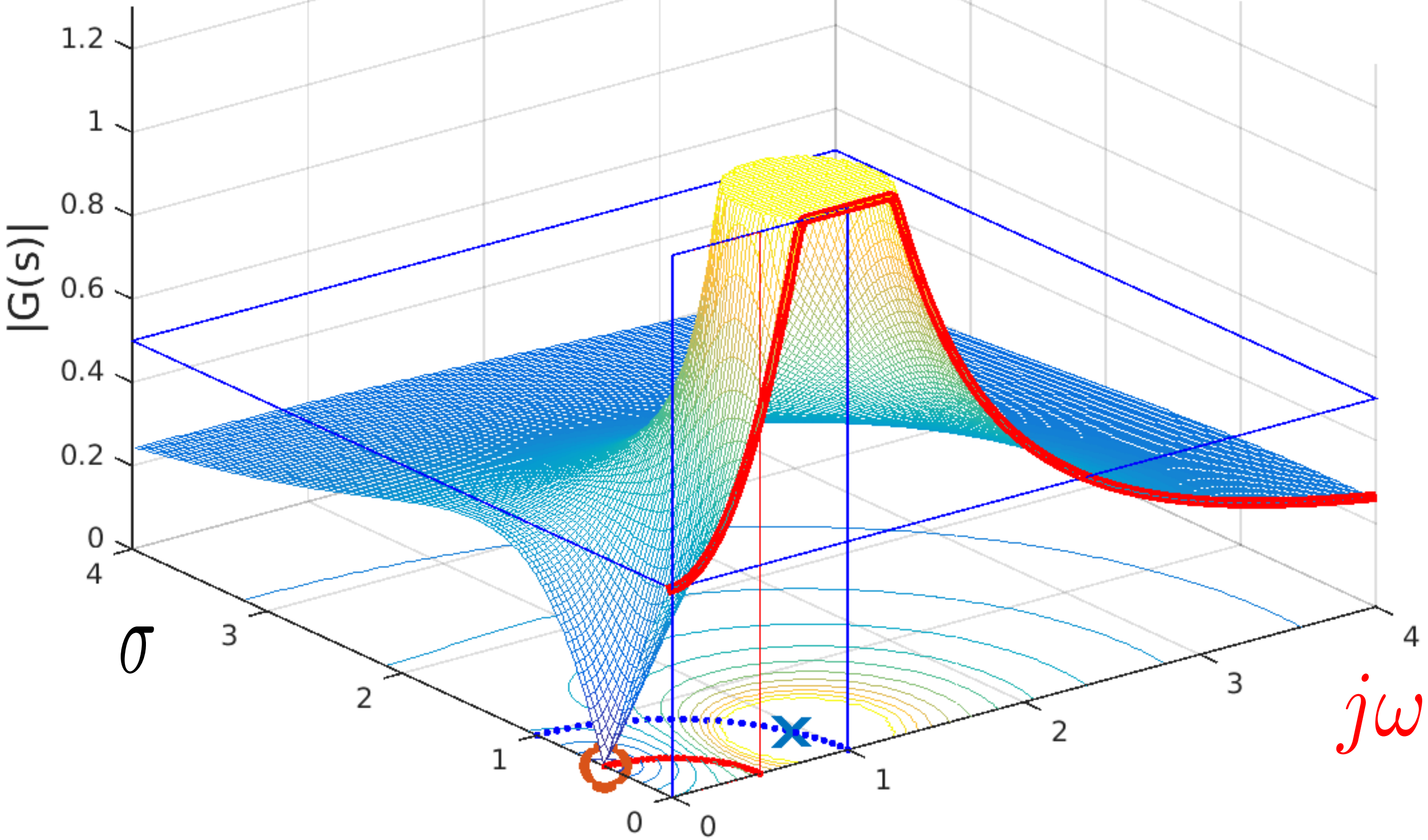
$$G = \frac{s + 0.5}{s^2 + 2 \cdot 0.34s + 1}$$

$\omega_n = 1$

$P = -0.34 \pm 0.94j$

$Z = -0.5j$

$\zeta = 0.34$



S-plane - plot 9

$$G = \frac{s + 0.5}{(s + 0.17 + 0.98j)(s + 0.17 - 0.98j)}$$

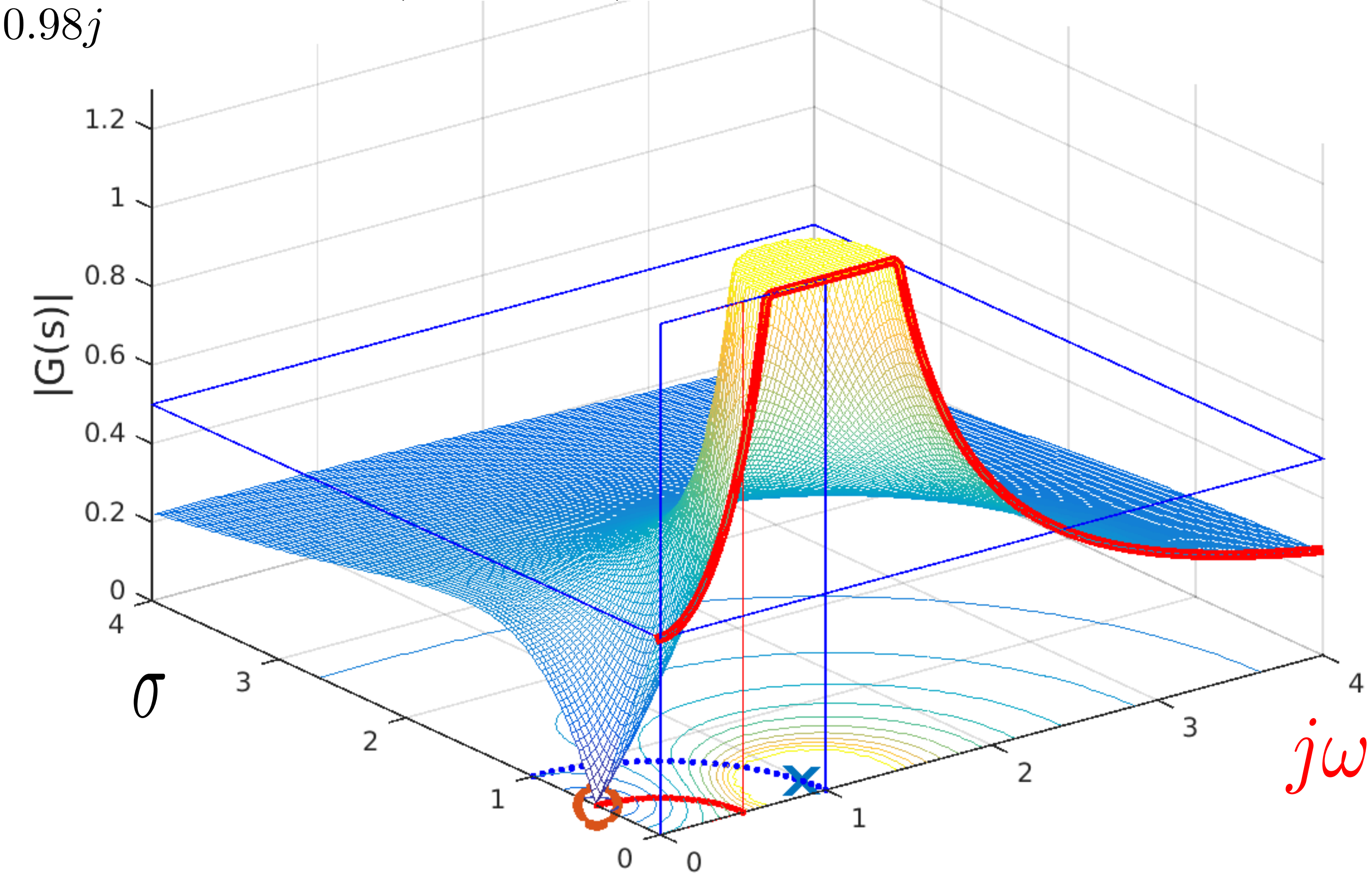
$$G = \frac{s + 0.5}{s^2 + 2 \cdot 0.17s + 1}$$

$\omega_n = 1$

$P = -0.17 \pm 0.98j$

$Z = -0.5j$

$\zeta = 0.17$



S-plane - plot 10

$$G = \frac{s+0.5}{(s+j)(s-j)}$$

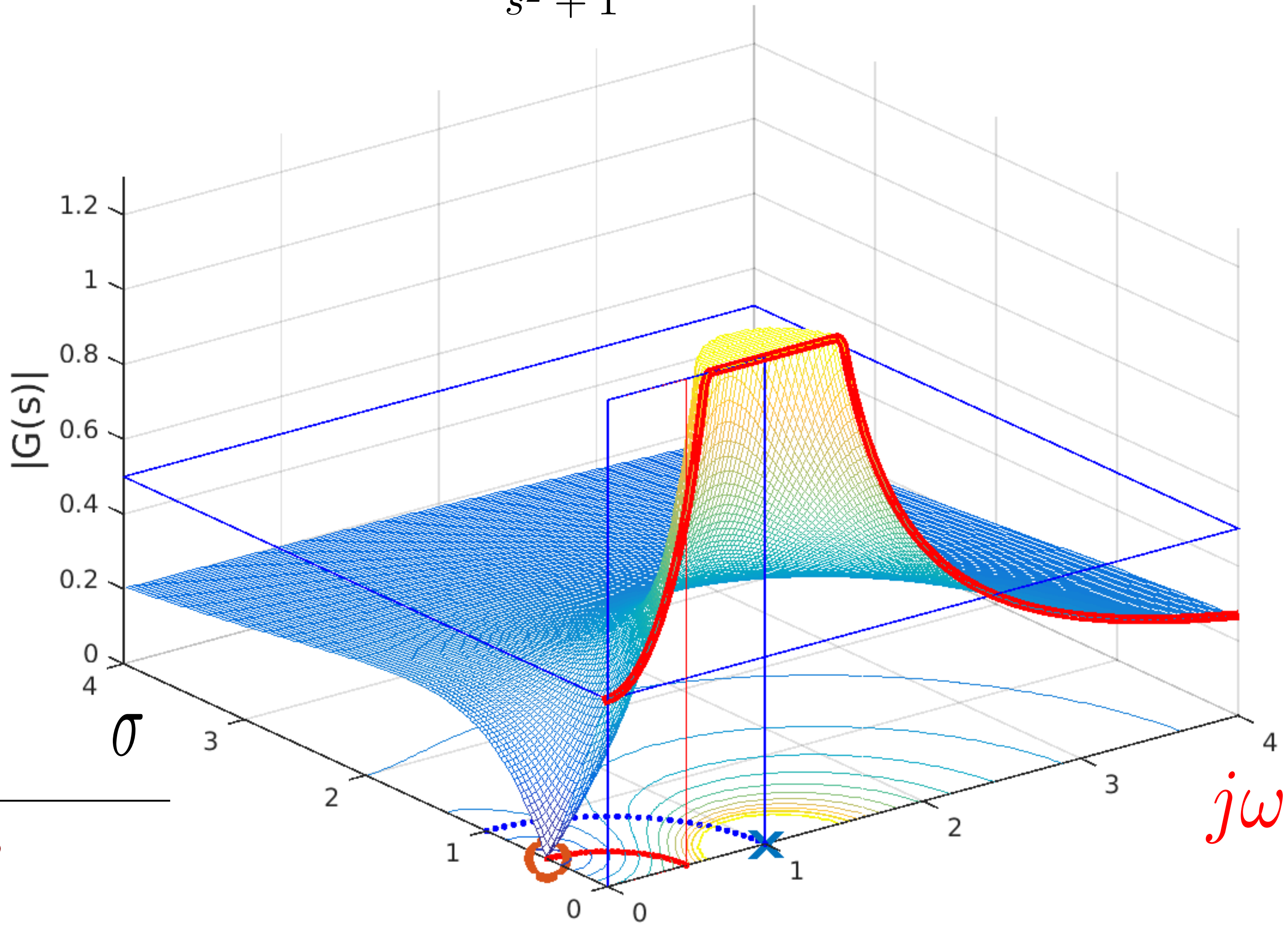
$$G = \frac{s+0.5}{s^2+1}$$

$\omega_n = 1$

$P = \pm j$

$Z = -0.5j$

$\zeta = 0$



Complex zeros?

The amplitude has negative peak at the natural frequency of the numerator

Bode plot – 2nd order

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s = j\omega$$

$$\omega \to 0 \Rightarrow G(\omega) \to 1\angle 0$$

$$\omega \to \infty \Rightarrow G(\omega) \to 0\angle - 180^o$$

At the natural frequency ω_n

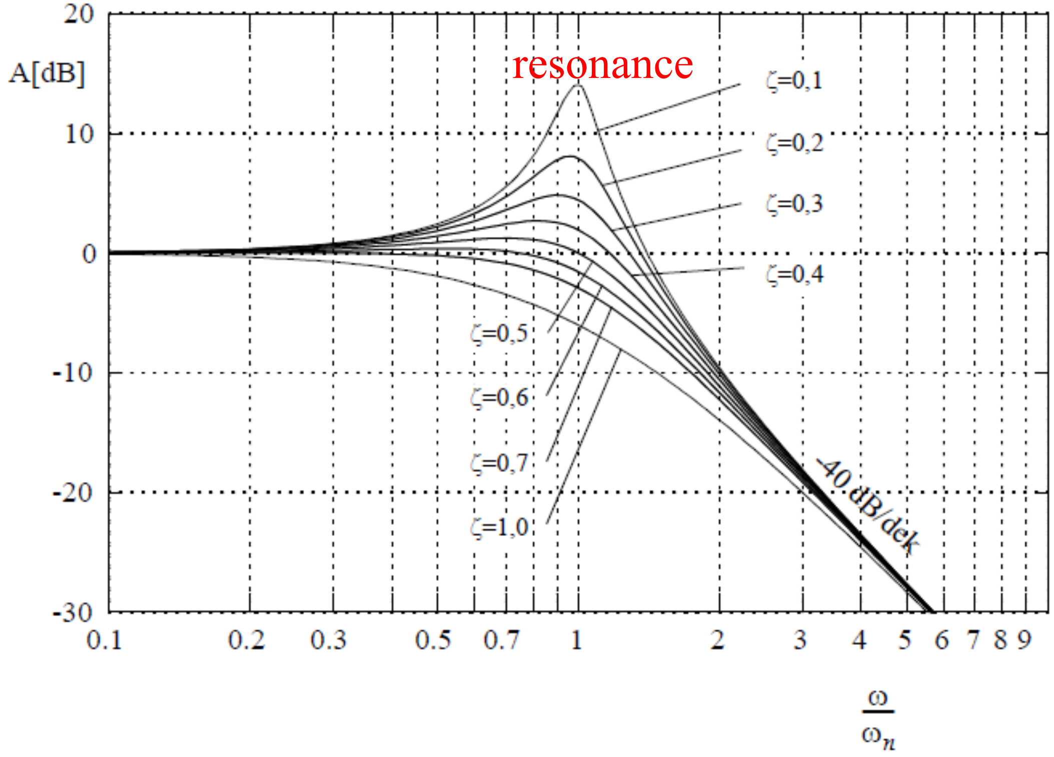
$$G(\omega_n) = \frac{\omega_n^2}{2\zeta\omega_n\omega_n j} = \frac{1}{2\zeta}\angle - 90^o$$

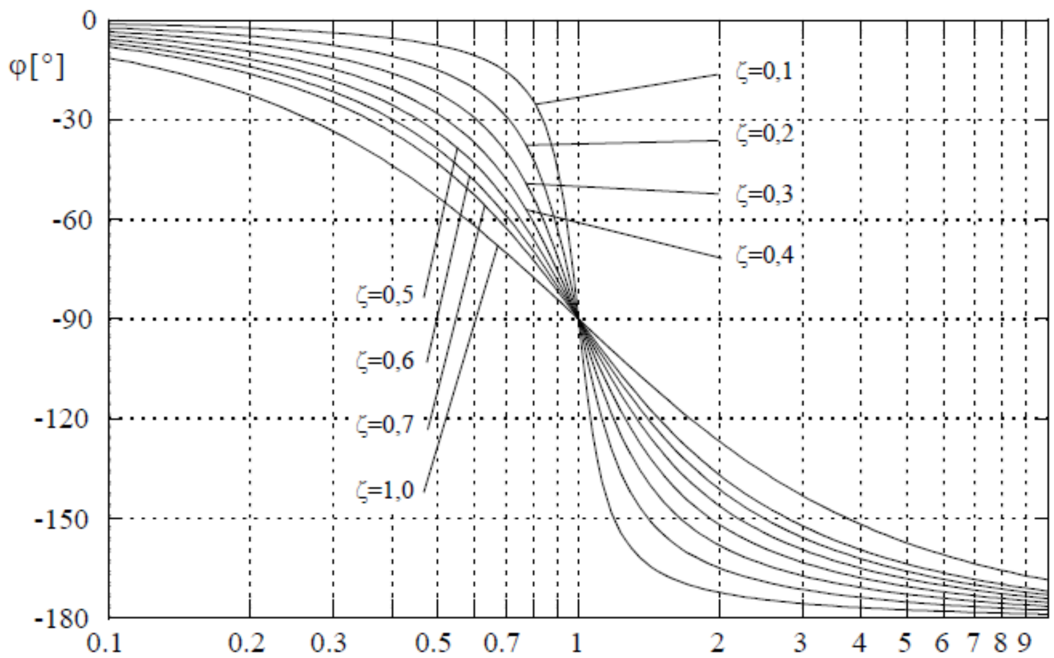
Question:

$$\zeta = 0.05 \Rightarrow \ |G(\omega_n)|_{dB} \ ?$$

$$\frac{1}{2 * 0.05} = 10 \Longrightarrow 20\ dB$$

34721





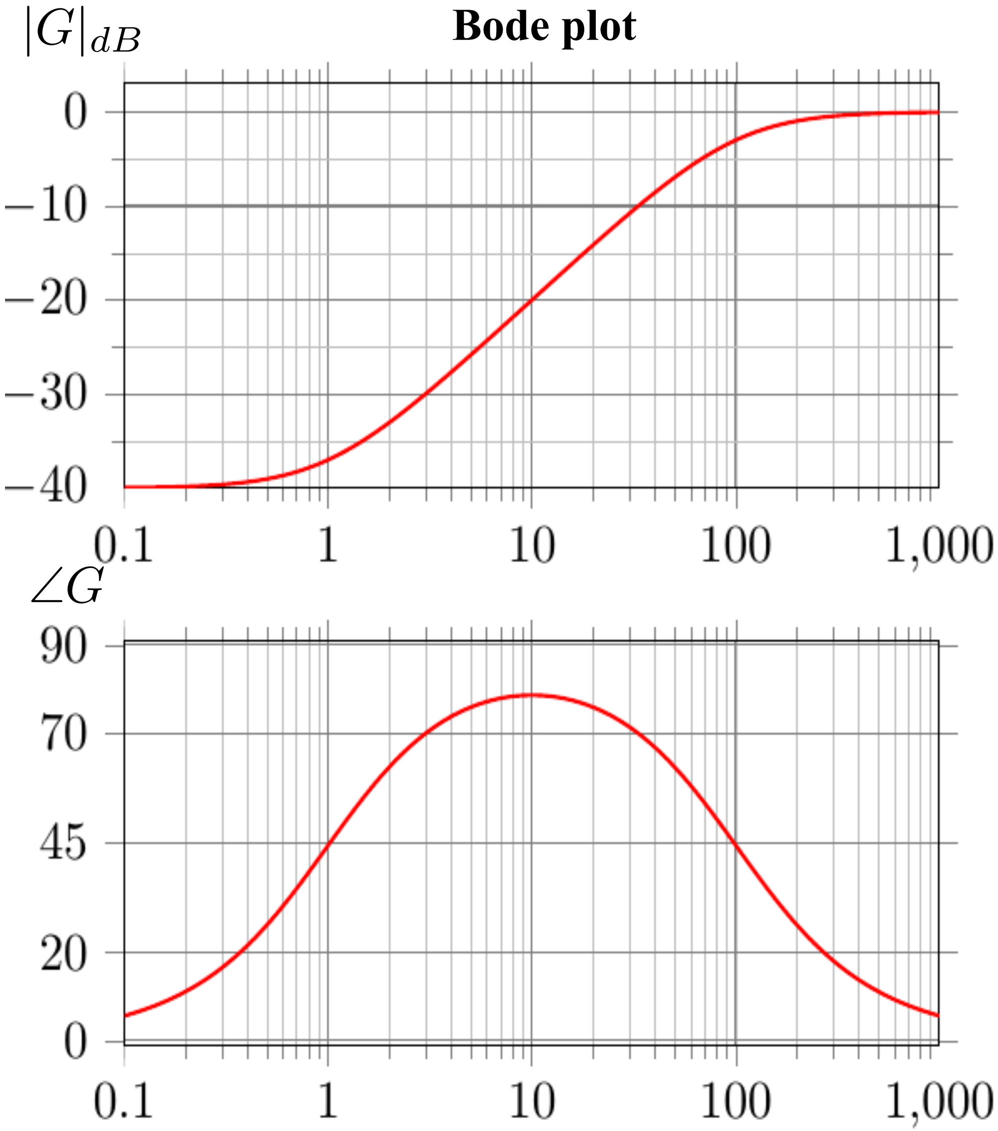
TFs with all poles and zeros in the LHP (or on the $j\omega$ axis) will be called

Minimum-phase systems

For these systems, there is a strict relation between the magnitude and the phase.

If the amplitude shape is known, the phase shape can be directly derived

Bode plot



Flat magnitude (0 dB/dec) → phase ≈ 0°

+20 dB/dec slope → phase → +90°

−20 dB/dec slope → phase → −90°

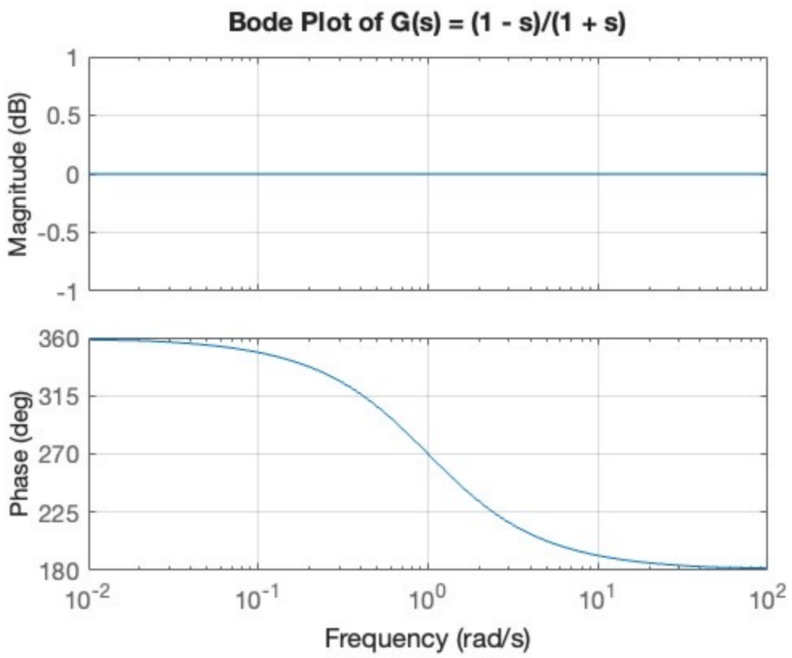
Non-minimum phase system:
At least one pole or zero point in the right half-plane

The **phase lag is larger** than expected from the magnitude plot.
The **step response initially moves in the wrong direction.**
Control design becomes **more difficult**.

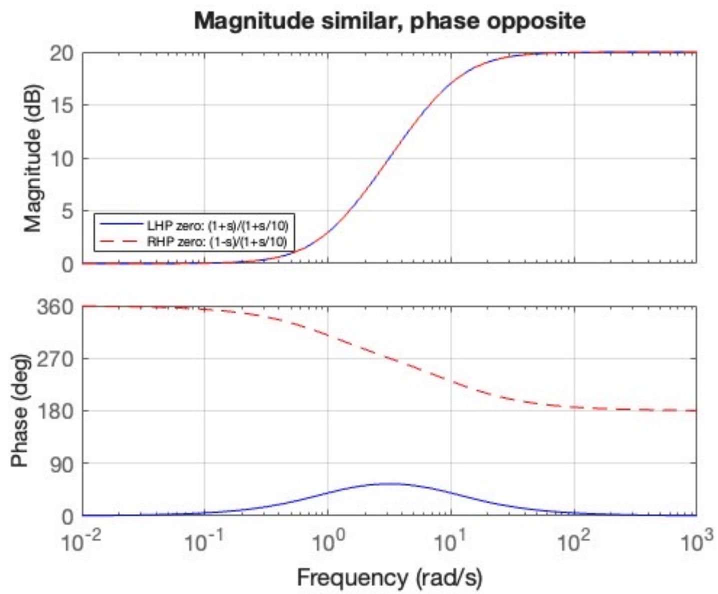
$$G(s) = \frac{1-s}{1+s}$$

This shows:

- magnitude looks like a normal zero
- phase goes the opposite direction



LHP zero vs RHP zero with the same pole

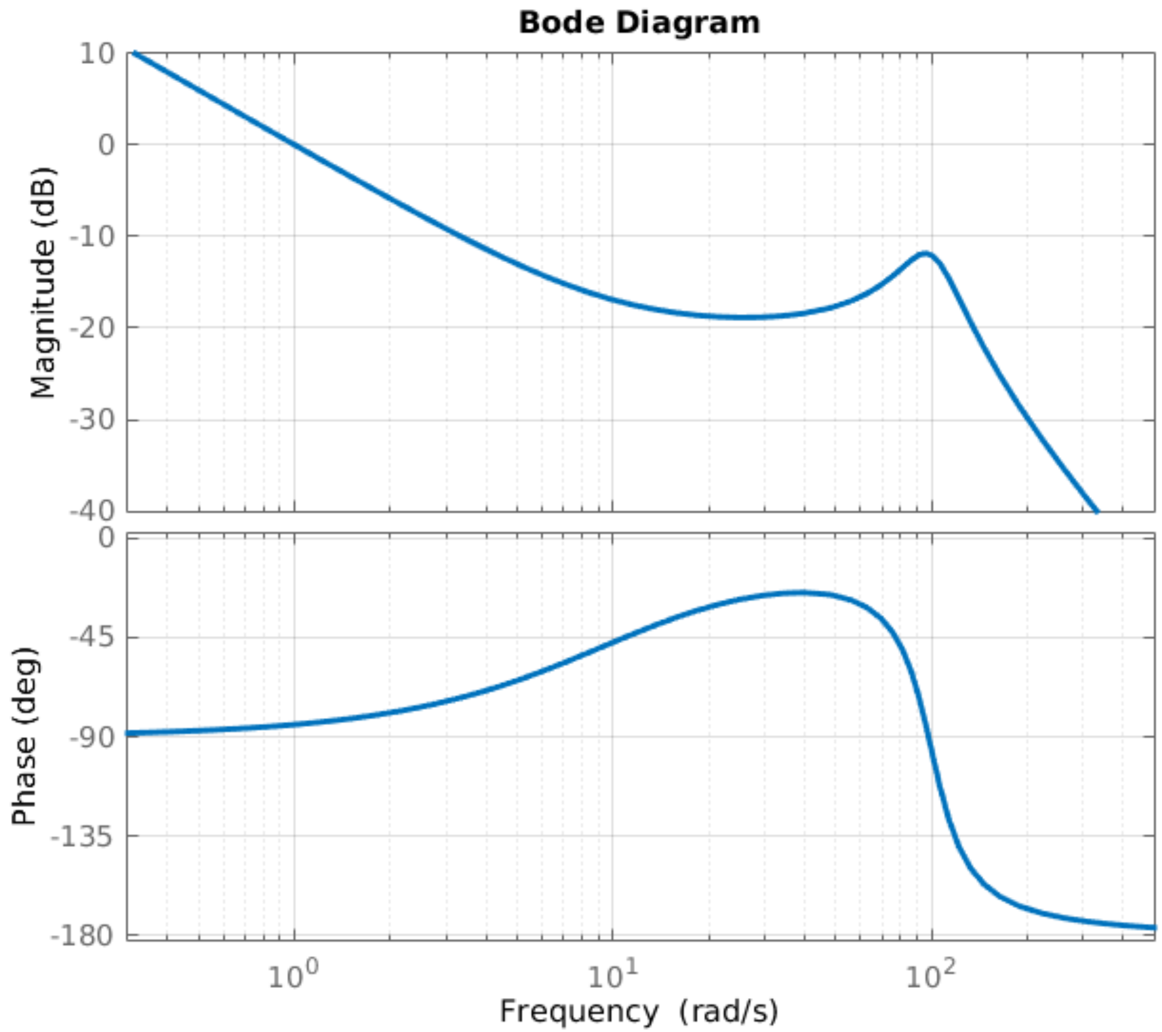


Control Questions

Given a system, a measurement gives the plot on the right.

What is the system's transfer function (approx.)?

$$G(s) = \frac{\cdots}{\cdots}$$

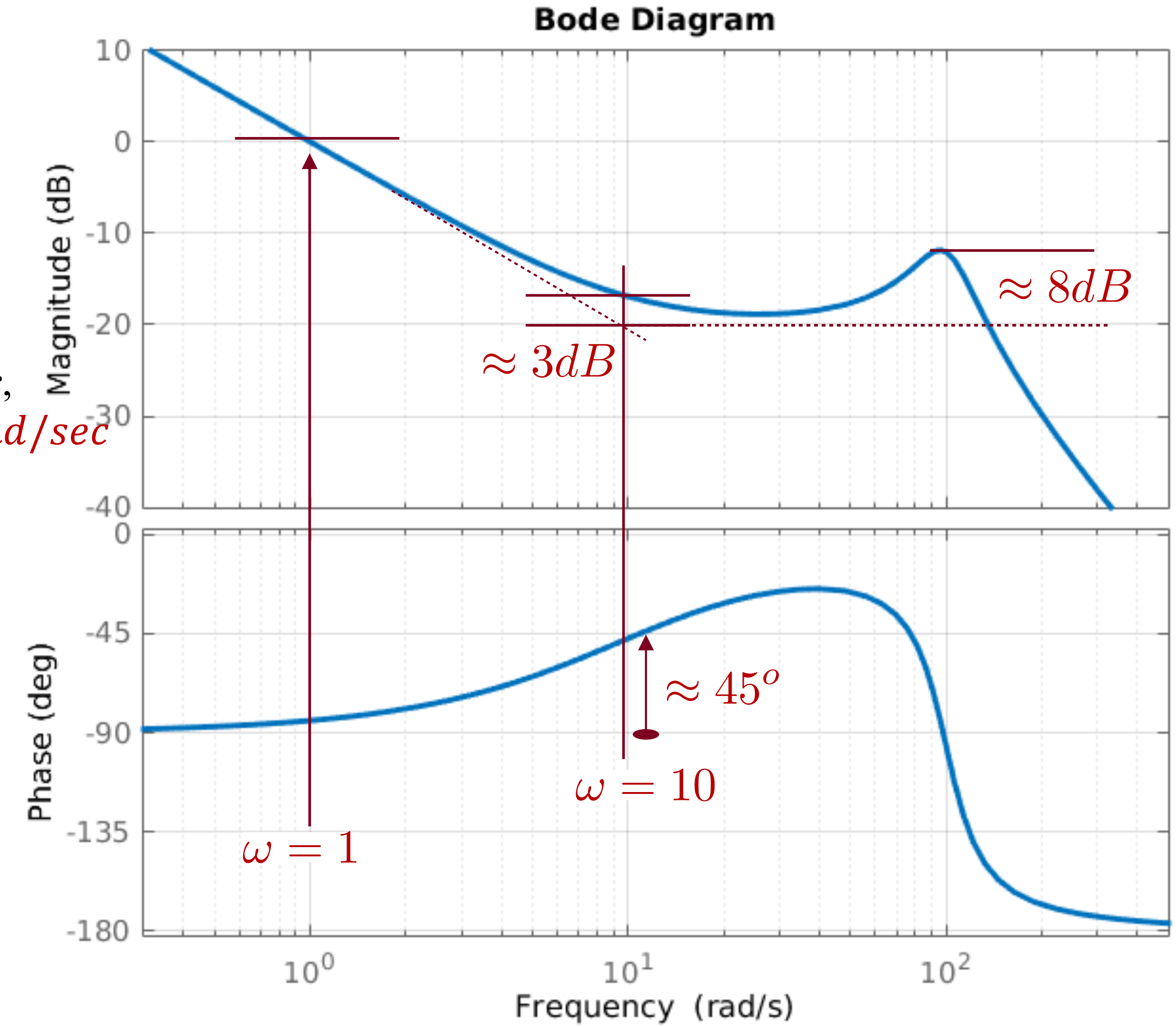


Control questions - Solution

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Transfer function (approx.)?

The plot starts with -20dB/dec and -90 degrees phase → **pole in s = 0**
There is something resembling a bend upwards at about 10 rad/sec
→ **real zero point** at s = -10
The amplitude peak and angle drop by approx. -180 degrees → **complex pole pair**, with resonance frequency on $\omega_n \approx 100\ rad/sec$ and peak is approx. 8 dB above the "flat" level → $\zeta \approx 0.2$



$$G(s) = \frac{k(s+10)}{s(s^2 + 40s + 10000)}$$

The constant k:

$$s = 1j \Rightarrow |G(j)| \approx 0dB$$

$$|G(s=j)| = \left|\frac{k(j+10)}{j(j^2+40j+10000)}\right| = 1 \Rightarrow \left|\frac{10k}{10000j}\right| = 1 \Rightarrow k = 1000$$

34721/34722 Linear Control Design 1
Spring Semester

Teachers: Silvia Tolu - Dimitrios Papageorgiou

Lesson 6/13 – Bode Plot and P-controller Design

Schedule

13:00 - 15:00

- Frequency analysis
- Bode plot - 1st and 2nd order
- **Stability**
- **Stability margins**

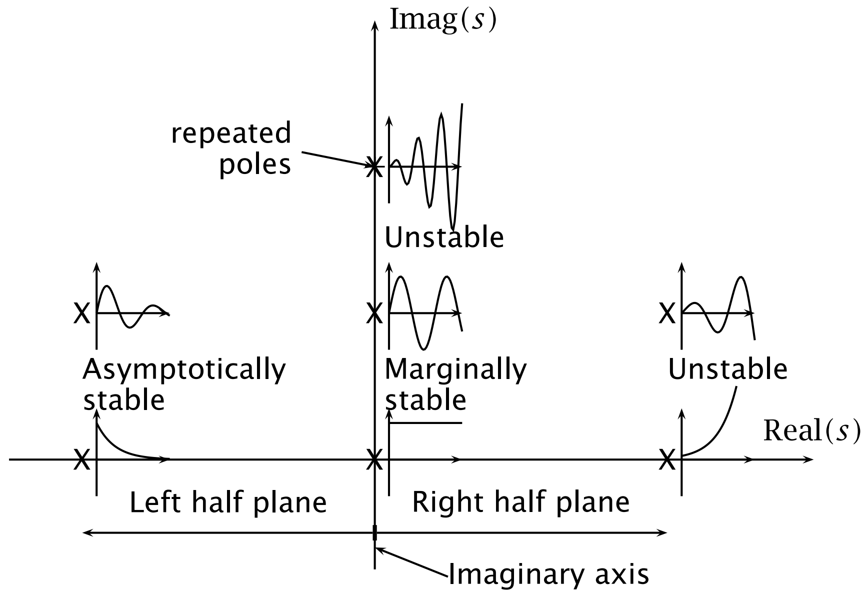
15:00 - 17:00

- Exercise **Day 6**

Stability

$$\int_0^\infty |g(t)|dt < \infty$$

- A system is asymptotically stable if **all** its poles have **negative real** parts.
- A system is unstable if any pole has a **positive real** part, or if there are any **repeated poles** on the imaginary axis. (➔ a term in the impulse response that blows up exponentially)
- A system is **marginally stable** if it has **one or more distinct poles on the imaginary axis**, and any remaining poles have negative real parts. (➔ terms in the impulse response that remain bounded, steps, or sinusoid)

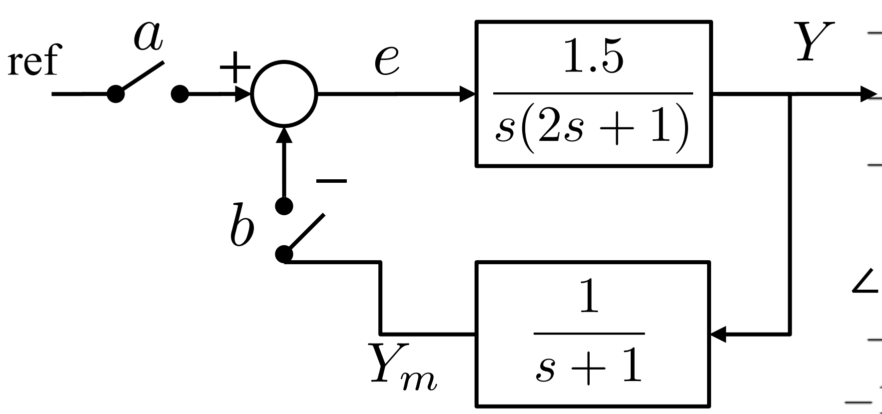


$$\int_0^T |g(t)|dt < A + BT \; for \; all \; T$$

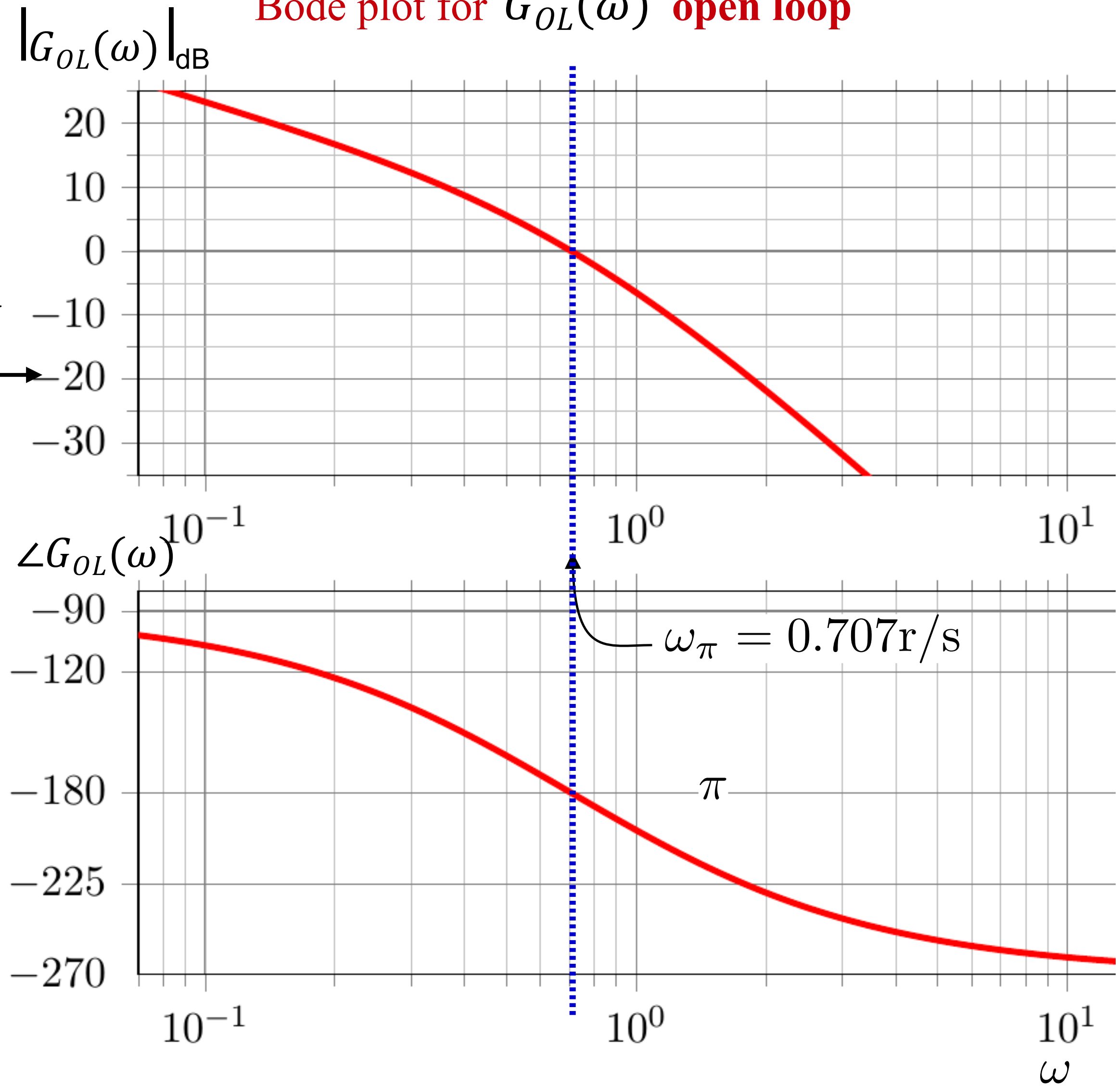
Stability

$$ref = \sin(\omega_\pi t) = \sin(0.707t)$$

$$G_{OL}(s) = \frac{1.5}{s(2s+1)(s+1)}$$



Bode plot for $G_{OL}(\omega)$ **open loop**



When a is closed and b is open

➔ e(t) = ref (t) - Y_m(t)

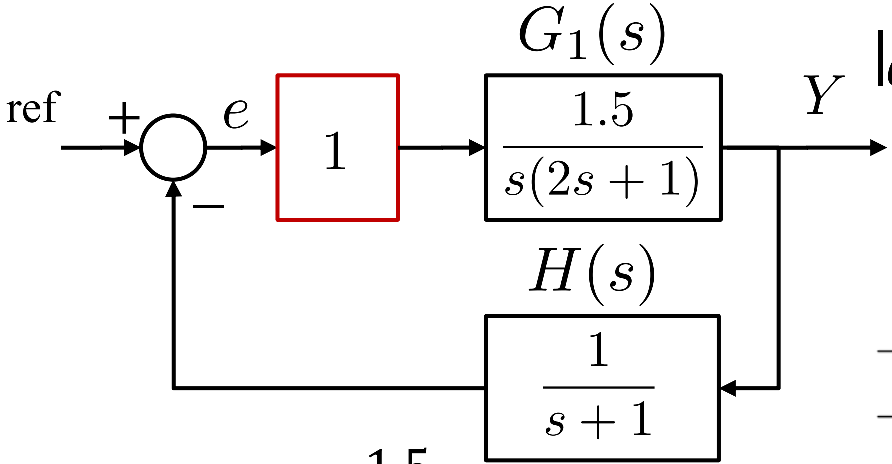
When b is closed and a is open

➔ e(t) = -Y_m(t)

Marginally stable

Critical point for stability $1\angle -180^o$

Stability



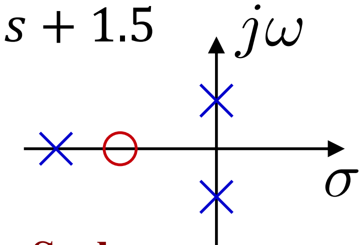
$$G_{OL}(s) = \frac{1.5}{s(2s+1)(s+1)}$$

Closed loop

$$G_{CL}(s) = \frac{G_1}{1 + G_1 H}$$

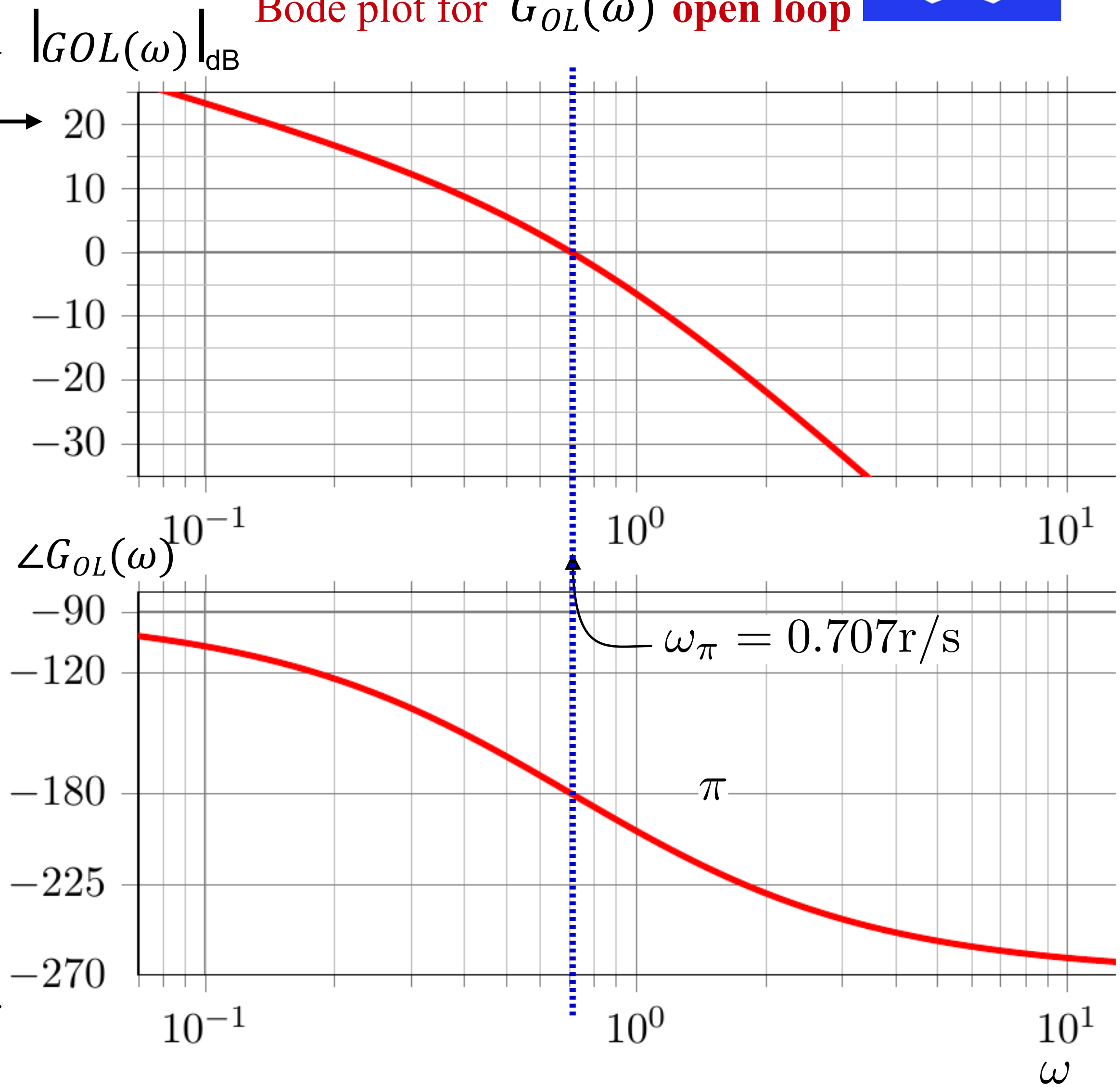
$$G_{CL}(s) = \frac{1.5\,(s+1)}{2s^3 + 3s^2 + s + 1.5}$$

$$\begin{aligned} p_1 &= -1.5 \\ p_2 &= +0.707j \\ p_3 &= -0.707j \end{aligned}$$



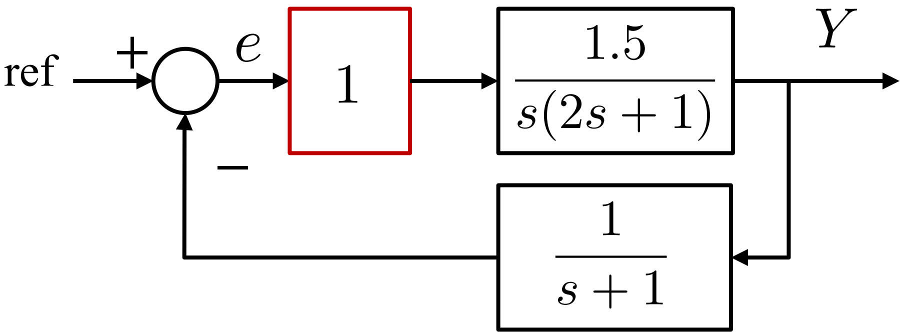
S-plane

Bode plot for $G_{OL}(\omega)$ **open loop**



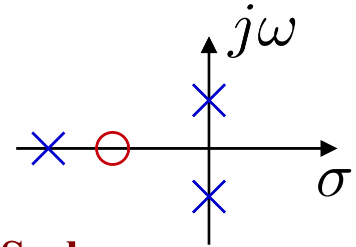
Critical point for stability $1\angle -180^o$
Closed loop

Closed-Loop Step Response

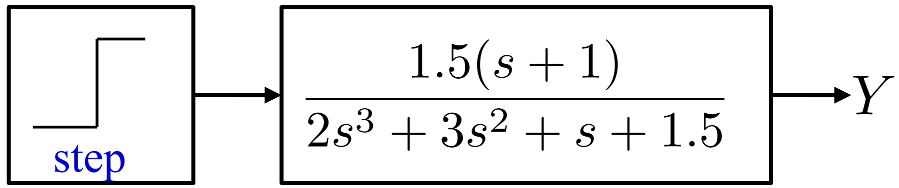


$$G_{CL}(s) = \frac{1.5\,(s+1)}{2s^3 + 3s^2 + s + 1.5}$$

$$\begin{aligned} p_1 &= -1.5 \\ p_2 &= +0.707j \\ p_3 &= -0.707j \end{aligned}$$

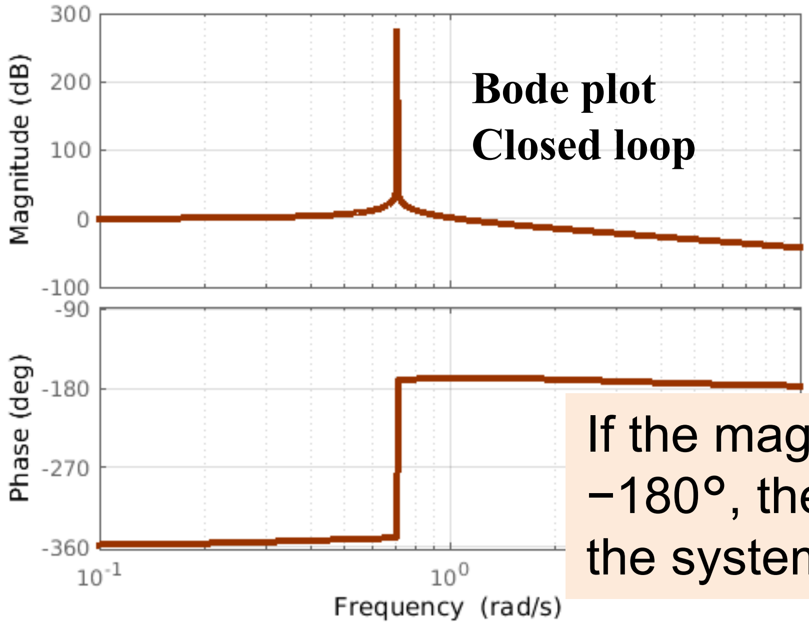


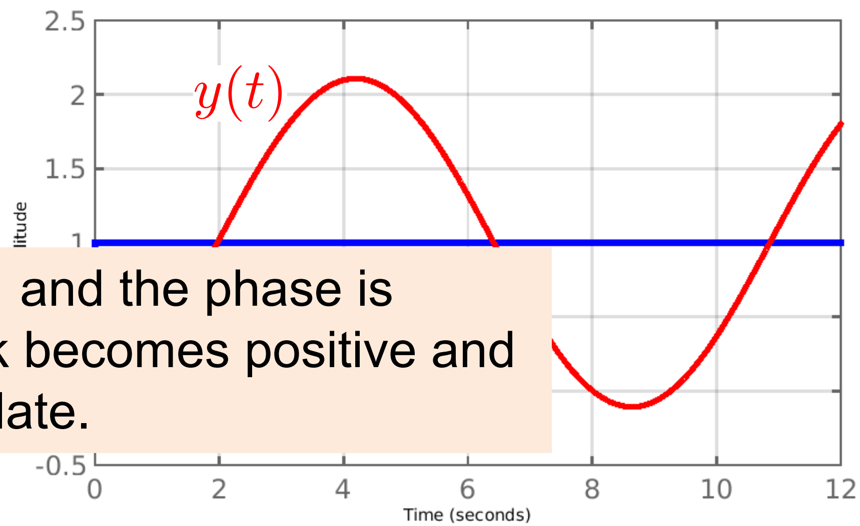
S-plane



$$y(t) = \mathcal{L}^{-1}\left\{\frac{1}{s}\,GCL(s)\right\}$$

$$y(t) = -1.1\cos(0.707t) + 0.2\sin(0.707t) + 1 + 0.1e^{-1.5t}$$

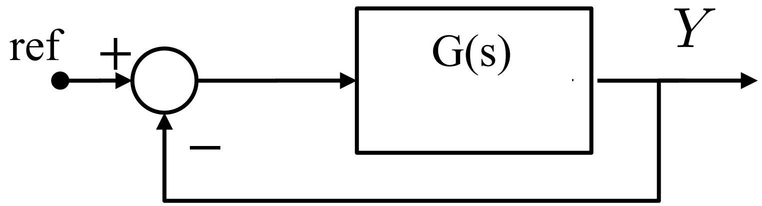




If the magnitude is 1 and the phase is −180°, the feedback becomes positive and the system will oscillate.

Stability Margins (OPEN LOOP)

The crossover frequency, $\boldsymbol{\omega_c}$ is often used to refer to the frequency at which the magnitude is unity or 0 dB.



Crossover frequency $\boldsymbol{\omega_c}$ ➔ $|G(j\omega_c)| = 0$ dB

>> ωc → faster response.

The phase margin (**PM**) is the amount by which the phase of $G(j\omega_c)$ exceeds -180 degrees when abs(G(ω)) = 1.
(it is most closely related to the damping ratio of the system)

Phase Margin (PM) $\gamma_M = 180° - |\gamma_c|$

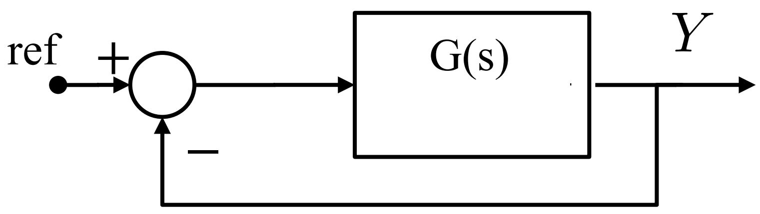
where $\gamma_c = \angle G(j\omega_c)$ (critical phase)

Larger PM → more damped and robust system.

The gain margin (**GM**) is the factor by which the gain can be increased (or decreased sometimes) before instability results.

Gain Margin (GM) $K_M = 0 - |G(j\omega_\pi)|$

Bode Stability Criterion

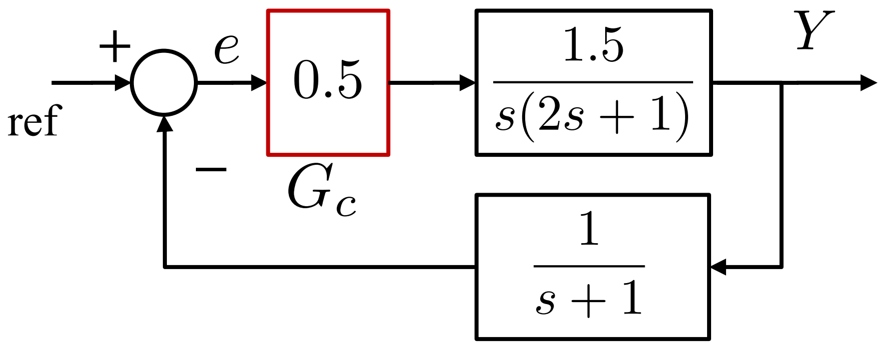


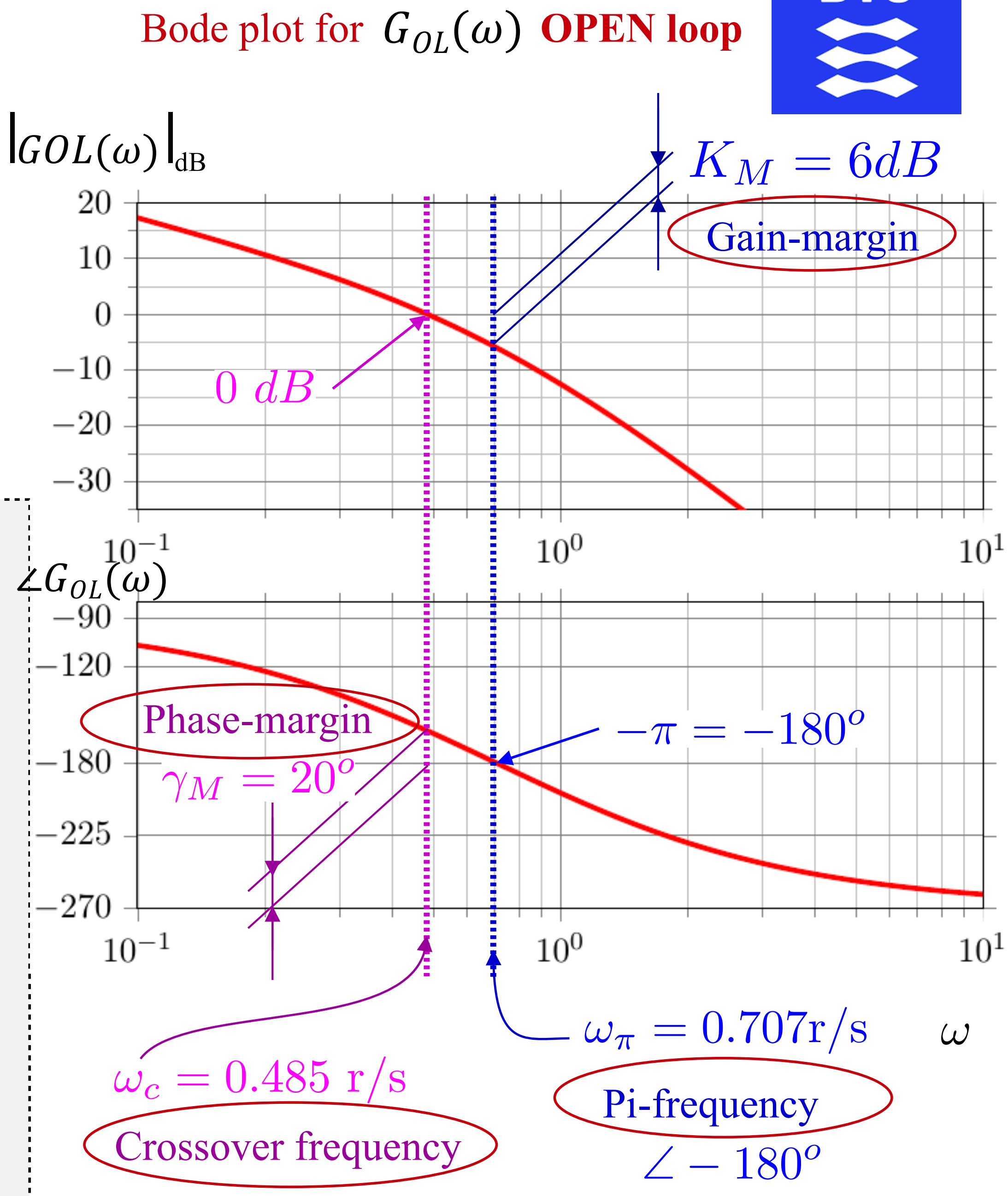
1. Stable System: both PM and GM should be positive. The greater the GM the more stable the system **(G(s)** does not have poles with real positive parts).

2. Marginal Stable System: both PM and GM are zero

1. Unstable System: If the PM or GM is negative

Stability Margins

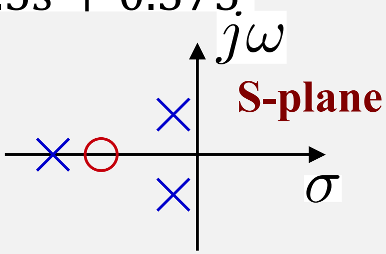


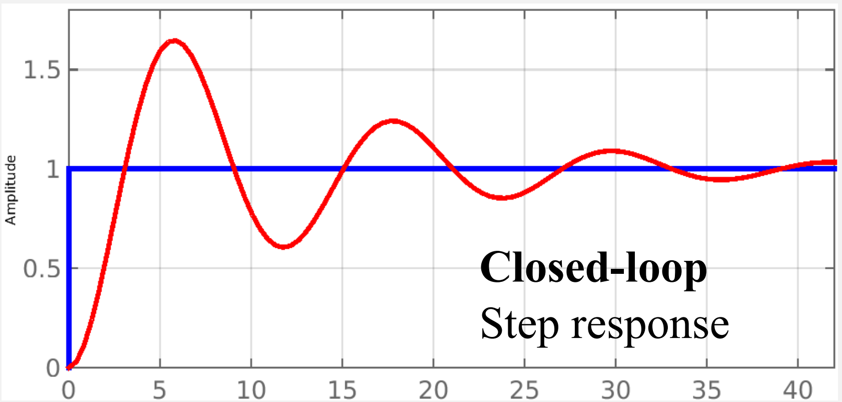


Closed loop

$$G_{CL}(s) = \frac{0.375\,(s+1)}{s^3 + 1.5s^2 + 0.5s + 0.375}$$

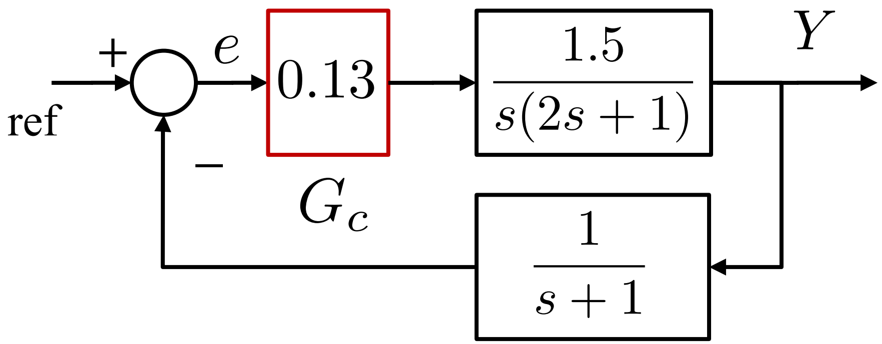
$$\begin{aligned} p_1 &= -1.34 \\ p_2 &= -0.08 + 0.5j \\ p_3 &= -0.08 - 0.5j \end{aligned}$$





LCD1 $|GOL| = 0$ dB

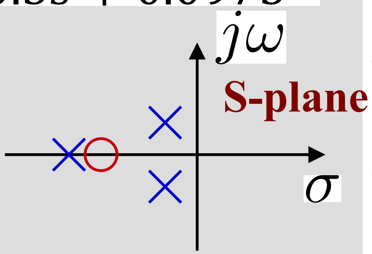
Stability Phase Margin

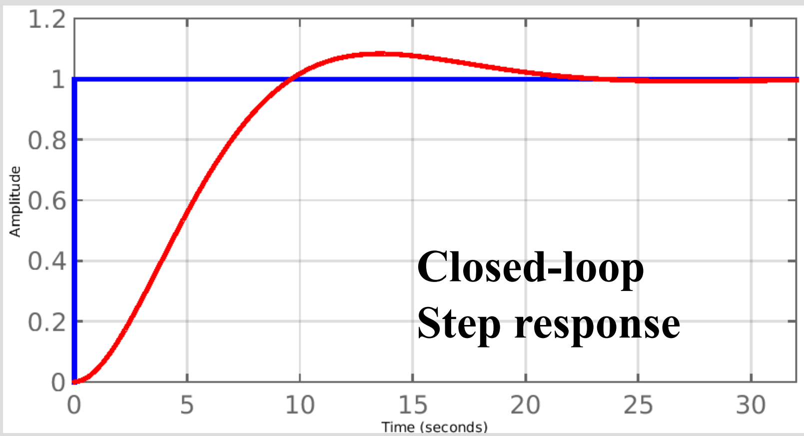


Closed loop

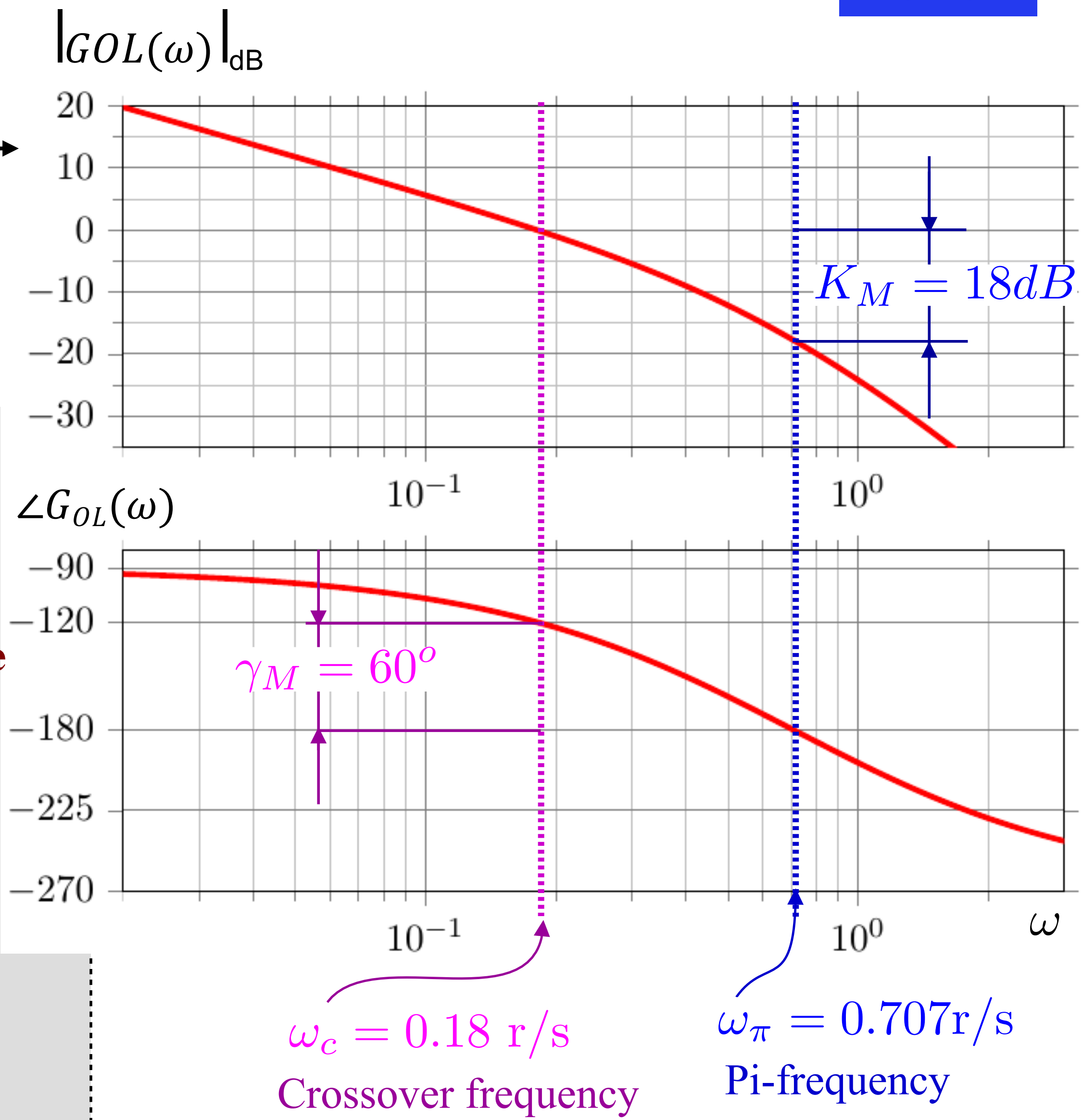
$$G_{CL}(s) = \frac{0.0975\,(s+1)}{s^3 + 1.5s^2 + 0.5s + 0.0975}$$

$$\begin{aligned} p_1 &= -1.14 \\ p_2 &= -0.18 + 0.23j \\ p_3 &= -0.18 - 0.23j \end{aligned}$$

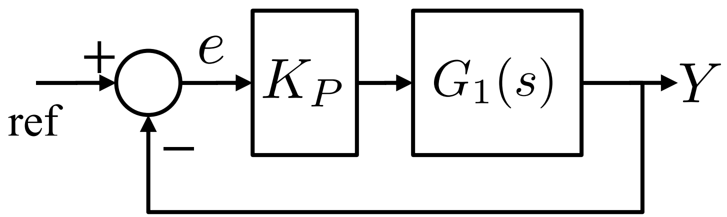




Bode plot for $G_{OL}(\omega)$ **OPEN loop**



Control Questions



Task 1

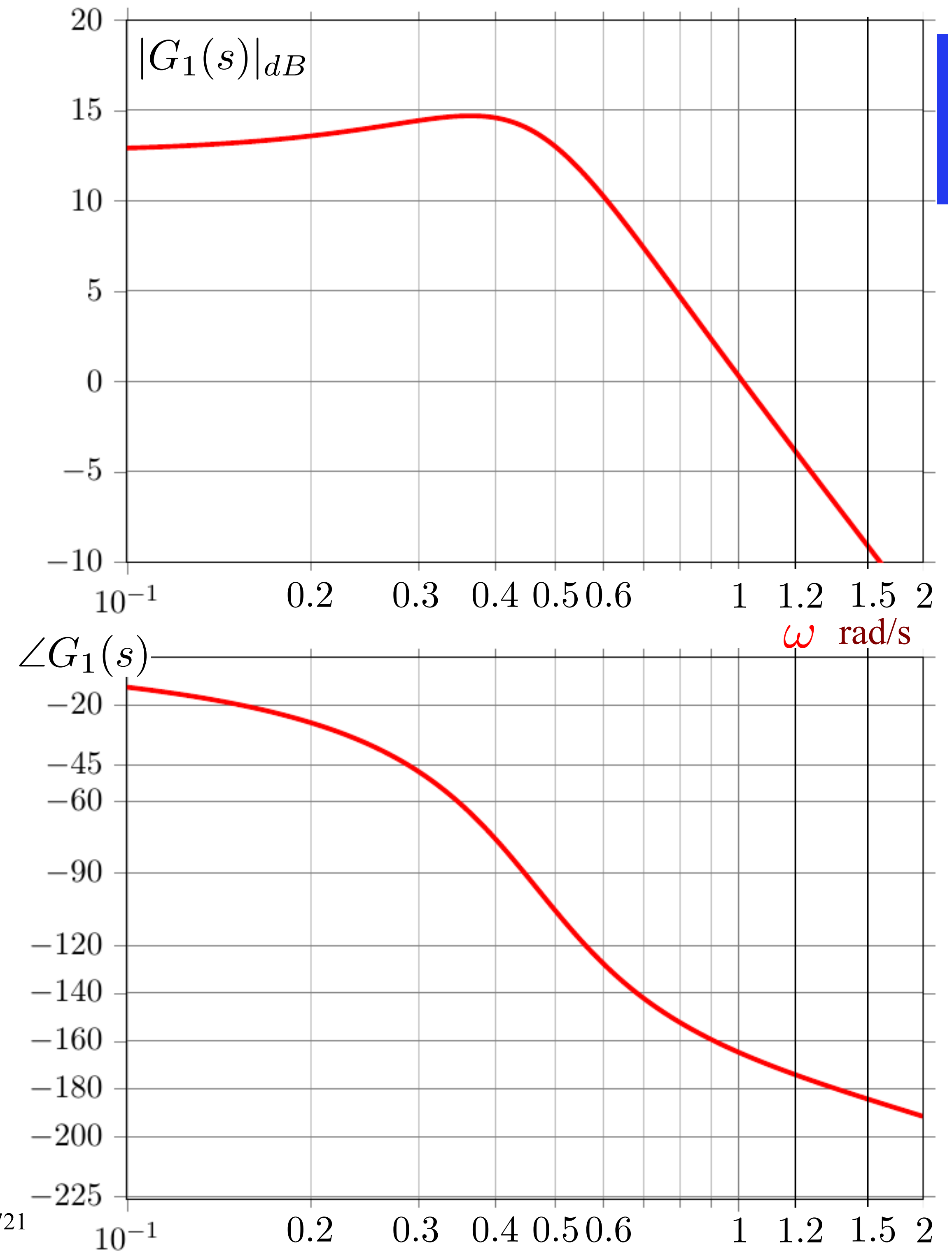
$K_P = 1$

Phase margin $\gamma_M = ?$

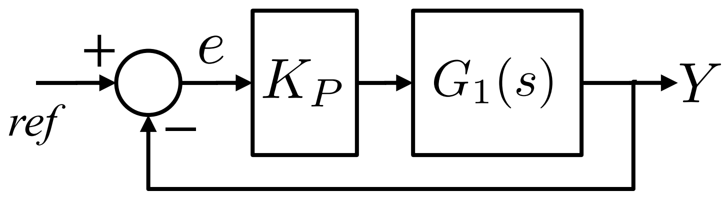
Gain-margin $K_M = ?$

Task 2

Find K_P so the open-loop phase margin is $\gamma_M = 60^o$



Control Questions - Solution

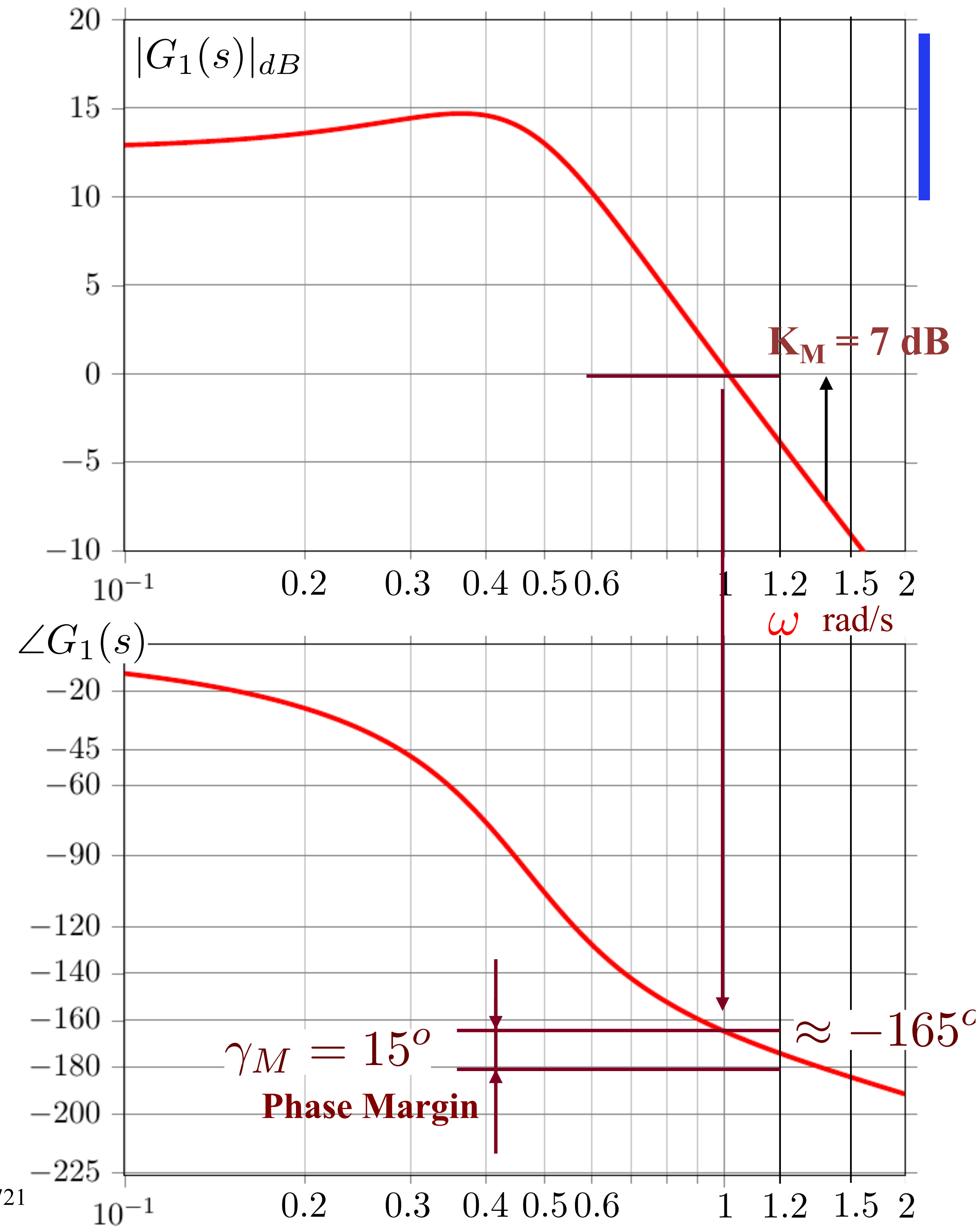


Task 1

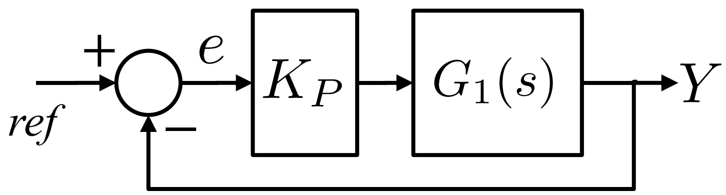
$$K_p = 1 \implies G_{OL} = G_1(s)$$

Phase margin $\quad \gamma_M = \ ?$

Gain-margin $\quad K_M = \ ?$



Control Questions - Solution



Task 2

Find K_P so open-loop phase margin $\gamma_M = 60^o$

$$\Rightarrow K_P = -12\ \text{dB}$$
$$\Rightarrow K_P = 10^{\frac{-12}{20}}$$
$$\Rightarrow K_P = 0.25$$